STRAVA FITNESS
My SQL Report

Github: Strava_Fitness

SQL WORKFLOW:

Tool used: **MySQL Work Bench**

```
--  ================================= --
-- |       **STRAVA FITNESS**        |
--  ================================= --
```

Database Initialization:

```
Create database Strava;
use Strava;
```

LOAD ALL THE DATASETS

1. dailyactivity

```
CREATE TABLE dailyactivity (
    Id BIGINT,
    ActivityDate DATE,
    TotalSteps INT,
    TotalDistance FLOAT,
    TrackerDistance FLOAT,
    LoggedActivitiesDistance FLOAT,
    VeryActiveDistance FLOAT,
    ModeratelyActiveDistance FLOAT,
    LightActiveDistance FLOAT,
    SedentaryActiveDistance FLOAT,
    VeryActiveMinutes INT,
    FairlyActiveMinutes INT,
    LightlyActiveMinutes INT,
    SedentaryMinutes INT,
    Calories INT
);
SET GLOBAL local_infile = 1;

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailyActivity_merged.csv"
INTO TABLE dailyactivity
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDate, @TotalSteps, @TotalDistance, @TrackerDistance,
@LoggedActivitiesDistance,
 @VeryActiveDistance, @ModeratelyActiveDistance, @LightActiveDistance,
@SedentaryActiveDistance,
 @VeryActiveMinutes, @FairlyActiveMinutes, @LightlyActiveMinutes,
@SedentaryMinutes, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityDate = STR_TO_DATE(@ActivityDate, '%c/%e/%Y'),  -- Handles
single-digit months/days
    TotalSteps = NULLIF(@TotalSteps, ''),
```

```
    TotalDistance = NULLIF(@TotalDistance, ''),
    TrackerDistance = NULLIF(@TrackerDistance, ''),
    LoggedActivitiesDistance = NULLIF(@LoggedActivitiesDistance, ''),
    VeryActiveDistance = NULLIF(@VeryActiveDistance, ''),
    ModeratelyActiveDistance = NULLIF(@ModeratelyActiveDistance, ''),
    LightActiveDistance = NULLIF(@LightActiveDistance, ''),
    SedentaryActiveDistance = NULLIF(@SedentaryActiveDistance, ''),
    VeryActiveMinutes = NULLIF(@VeryActiveMinutes, ''),
    FairlyActiveMinutes = NULLIF(@FairlyActiveMinutes, ''),
    LightlyActiveMinutes = NULLIF(@LightlyActiveMinutes, ''),
    SedentaryMinutes = NULLIF(@SedentaryMinutes, ''),
    Calories = NULLIF(@Calories, '');
```

2. dailycalories:

```
CREATE TABLE dailycalories (
    Id BIGINT,
    ActivityDay DATE,
    Calories INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailyCalories_merged.csv"
INTO TABLE dailycalories
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDay, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityDay = STR_TO_DATE(@ActivityDay, '%c/%e/%Y'),
    Calories = NULLIF(@Calories, '');
```

3. dailyintensities

```
CREATE TABLE dailyintensities (
    Id BIGINT,
    ActivityDay DATE,
    SedentaryMinutes INT,
    LightlyActiveMinutes INT,
    FairlyActiveMinutes INT,
    VeryActiveMinutes INT,
    SedentaryActiveDistance int,
      LightActiveDistance double,
    ModeratelyActiveDistance double,
    VeryActiveDistance double
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailyIntensities_merged.csv"
INTO TABLE dailyintensities
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDay, @SedentaryMinutes, @LightlyActiveMinutes,
@FairlyActiveMinutes, @VeryActiveMinutes, @SedentaryActiveDistance,
@LightActiveDistance,@ModeratelyActiveDistance,@VeryActiveDistance)
SET
```

```
    Id = NULLIF(@Id, ''),
    ActivityDay = STR_TO_DATE(@ActivityDay, '%c/%e/%Y'),
    SedentaryMinutes = NULLIF(@SedentaryMinutes, ''),
    LightlyActiveMinutes = NULLIF(@LightlyActiveMinutes, ''),
    FairlyActiveMinutes = NULLIF(@FairlyActiveMinutes, ''),
    VeryActiveMinutes = NULLIF(@VeryActiveMinutes, ''),
    SedentaryActiveDistance = NULLIF(@SedentaryActiveDistance, ''),
    LightActiveDistance = NULLIF(@LightActiveDistance, ''),
    ModeratelyActiveDistance = NULLIF(@ModeratelyActiveDistance, ''),
    VeryActiveDistance = NULLIF(@VeryActiveDistance, '');
```

4. Dailysteps

```
CREATE TABLE dailysteps (
    Id BIGINT,
    ActivityDay DATE,
    StepTotal INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/dailySteps_merged.csv"
INTO TABLE dailysteps
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityDay, @StepTotal)
SET
    Id = NULLIF(@Id, ''),
    ActivityDay = STR_TO_DATE(@ActivityDay, '%c/%e/%Y'),
    StepTotal = NULLIF(@StepTotal, '');
```

5. Heartrateseconds

```
CREATE TABLE heartrateseconds (
    Id BIGINT,
    Time DATETIME,
    Value INT);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/heartrate_seconds_merged.csv"
INTO TABLE heartrateseconds
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @Time, @Value)
SET
    Id = NULLIF(@Id, ''),
    Time = STR_TO_DATE(@Time, '%m/%d/%Y %r'),
    Value = NULLIF(@Value, '');
```

6. Hourlycalories

```
CREATE TABLE hourlycalories (
    Id BIGINT,
    ActivityHour DATETIME,
    Calories INT
```

```
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/hourlyCalories_merged.csv"
INTO TABLE hourlycalories
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    Calories = NULLIF(@Calories, '');
```

7. Hourlyintensities

```
 CREATE TABLE hourlyintensities (
    Id BIGINT,
    ActivityHour DATETIME,
    TotalIntensity INT,
    AverageIntensity DOUBLE
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/hourlyIntensities_merged.csv"
INTO TABLE hourlyintensities
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour, @TotalIntensity, @AverageIntensity)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    TotalIntensity = NULLIF(@TotalIntensity, ''),
    AverageIntensity = NULLIF(@AverageIntensity, '');
```

8. hourlysteps

```
CREATE TABLE hourlysteps (
    Id BIGINT,
    ActivityHour DATETIME,
    StepTotal INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/hourlySteps_merged.csv"
INTO TABLE hourlysteps
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour, @StepTotal)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    StepTotal = NULLIF(@StepTotal, '');
```

9. Minutecaloriesnarrow

```
 CREATE TABLE minutecaloriesnarrow (
    Id BIGINT,
    ActivityMinute DATETIME,
    Calories FLOAT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteCaloriesNarrow_merged.csv"
INTO TABLE minutecaloriesnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @Calories)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    Calories = NULLIF(@Calories, '');
```

10. Minutecalorieswide

```
CREATE TABLE minutecalorieswide (
    Id BIGINT,
    ActivityMinute DATETIME,
    Calories00 DOUBLE,
    Calories01 DOUBLE,
    Calories02 DOUBLE,
    Calories03 DOUBLE,
    Calories04 DOUBLE,
    Calories05 DOUBLE,
    Calories06 DOUBLE,
    Calories07 DOUBLE,
    Calories08 DOUBLE,
    Calories09 DOUBLE,
    Calories10 DOUBLE,
    Calories11 DOUBLE,
    Calories12 DOUBLE,
    Calories13 DOUBLE,
    Calories14 DOUBLE,
    Calories15 DOUBLE,
    Calories16 DOUBLE,
    Calories17 DOUBLE,
    Calories18 DOUBLE,
    Calories19 DOUBLE,
    Calories20 DOUBLE,
    Calories21 DOUBLE,
    Calories22 DOUBLE,
    Calories23 DOUBLE,
    Calories24 DOUBLE,
    Calories25 DOUBLE,
    Calories26 DOUBLE,
    Calories27 DOUBLE,
    Calories28 DOUBLE,
    Calories29 DOUBLE,
    Calories30 DOUBLE,
```

```
    Calories31 DOUBLE,
    Calories32 DOUBLE,
    Calories33 DOUBLE,
    Calories34 DOUBLE,
    Calories35 DOUBLE,
    Calories36 DOUBLE,
    Calories37 DOUBLE,
    Calories38 DOUBLE,
    Calories39 DOUBLE,
    Calories40 DOUBLE,
    Calories41 DOUBLE,
    Calories42 DOUBLE,
    Calories43 DOUBLE,
    Calories44 DOUBLE,
    Calories45 DOUBLE,
    Calories46 DOUBLE,
    Calories47 DOUBLE,
    Calories48 DOUBLE,
    Calories49 DOUBLE,
    Calories50 DOUBLE,
    Calories51 DOUBLE,
    Calories52 DOUBLE,
    Calories53 DOUBLE,
    Calories54 DOUBLE,
    Calories55 DOUBLE,
    Calories56 DOUBLE,
    Calories57 DOUBLE,
    Calories58 DOUBLE,
    Calories59 DOUBLE
);
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteCaloriesWide_merged.csv"
INTO TABLE minutecalorieswide
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS

(@Id, @ActivityMinute,
 @Calories00, @Calories01, @Calories02, @Calories03, @Calories04,
@Calories05, @Calories06, @Calories07, @Calories08, @Calories09,
 @Calories10, @Calories11, @Calories12, @Calories13, @Calories14,
@Calories15, @Calories16, @Calories17, @Calories18, @Calories19,
 @Calories20, @Calories21, @Calories22, @Calories23, @Calories24,
@Calories25, @Calories26, @Calories27, @Calories28, @Calories29,
 @Calories30, @Calories31, @Calories32, @Calories33, @Calories34,
@Calories35, @Calories36, @Calories37, @Calories38, @Calories39,
 @Calories40, @Calories41, @Calories42, @Calories43, @Calories44,
@Calories45, @Calories46, @Calories47, @Calories48, @Calories49,
 @Calories50, @Calories51, @Calories52, @Calories53, @Calories54,
@Calories55, @Calories56, @Calories57, @Calories58, @Calories59)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    Calories00 = NULLIF(@Calories00, ''),
    Calories01 = NULLIF(@Calories01, ''),
    Calories02 = NULLIF(@Calories02, ''),
    Calories03 = NULLIF(@Calories03, ''),
```

```
    Calories04 = NULLIF(@Calories04, ''),
    Calories05 = NULLIF(@Calories05, ''),
    Calories06 = NULLIF(@Calories06, ''),
    Calories07 = NULLIF(@Calories07, ''),
    Calories08 = NULLIF(@Calories08, ''),
    Calories09 = NULLIF(@Calories09, ''),
    Calories10 = NULLIF(@Calories10, ''),
    Calories11 = NULLIF(@Calories11, ''),
    Calories12 = NULLIF(@Calories12, ''),
    Calories13 = NULLIF(@Calories13, ''),
    Calories14 = NULLIF(@Calories14, ''),
    Calories15 = NULLIF(@Calories15, ''),
    Calories16 = NULLIF(@Calories16, ''),
    Calories17 = NULLIF(@Calories17, ''),
    Calories18 = NULLIF(@Calories18, ''),
    Calories19 = NULLIF(@Calories19, ''),
    Calories20 = NULLIF(@Calories20, ''),
    Calories21 = NULLIF(@Calories21, ''),
    Calories22 = NULLIF(@Calories22, ''),
    Calories23 = NULLIF(@Calories23, ''),
    Calories24 = NULLIF(@Calories24, ''),
    Calories25 = NULLIF(@Calories25, ''),
    Calories26 = NULLIF(@Calories26, ''),
    Calories27 = NULLIF(@Calories27, ''),
    Calories28 = NULLIF(@Calories28, ''),
    Calories29 = NULLIF(@Calories29, ''),
    Calories30 = NULLIF(@Calories30, ''),
    Calories31 = NULLIF(@Calories31, ''),
    Calories32 = NULLIF(@Calories32, ''),
    Calories33 = NULLIF(@Calories33, ''),
    Calories34 = NULLIF(@Calories34, ''),
    Calories35 = NULLIF(@Calories35, ''),
    Calories36 = NULLIF(@Calories36, ''),
    Calories37 = NULLIF(@Calories37, ''),
    Calories38 = NULLIF(@Calories38, ''),
    Calories39 = NULLIF(@Calories39, ''),
    Calories40 = NULLIF(@Calories40, ''),
    Calories41 = NULLIF(@Calories41, ''),
    Calories42 = NULLIF(@Calories42, ''),
    Calories43 = NULLIF(@Calories43, ''),
    Calories44 = NULLIF(@Calories44, ''),
    Calories45 = NULLIF(@Calories45, ''),
    Calories46 = NULLIF(@Calories46, ''),
    Calories47 = NULLIF(@Calories47, ''),
    Calories48 = NULLIF(@Calories48, ''),
    Calories49 = NULLIF(@Calories49, ''),
    Calories50 = NULLIF(@Calories50, ''),
    Calories51 = NULLIF(@Calories51, ''),
    Calories52 = NULLIF(@Calories52, ''),
    Calories53 = NULLIF(@Calories53, ''),
    Calories54 = NULLIF(@Calories54, ''),
    Calories55 = NULLIF(@Calories55, ''),
    Calories56 = NULLIF(@Calories56, ''),
    Calories57 = NULLIF(@Calories57, ''),
    Calories58 = NULLIF(@Calories58, ''),
    Calories59 = NULLIF(@Calories59, '');
```

11. minuteintensitiesnarrow

```
CREATE TABLE minuteintensitiesnarrow (
    Id BIGINT,
    ActivityMinute DATETIME,
    Intensity INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteIntensitiesNarrow_merged.csv"
INTO TABLE minuteintensitiesnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @Intensity)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    Intensity = NULLIF(@Intensity, '');
```

12. ***minuteintensitieswide***

```
CREATE TABLE minuteintensitieswide (
    Id BIGINT,
    ActivityHour DATETIME,
    Intensity00 INT,
    Intensity01 INT,
    Intensity02 INT,
    Intensity03 INT,
    Intensity04 INT,
    Intensity05 INT,
    Intensity06 INT,
    Intensity07 INT,
    Intensity08 INT,
    Intensity09 INT,
    Intensity10 INT,
    Intensity11 INT,
    Intensity12 INT,
    Intensity13 INT,
    Intensity14 INT,
    Intensity15 INT,
    Intensity16 INT,
    Intensity17 INT,
    Intensity18 INT,
    Intensity19 INT,
    Intensity20 INT,
    Intensity21 INT,
    Intensity22 INT,
    Intensity23 INT,
    Intensity24 INT,
    Intensity25 INT,
    Intensity26 INT,
    Intensity27 INT,
    Intensity28 INT,
    Intensity29 INT,
    Intensity30 INT,
    Intensity31 INT,
    Intensity32 INT,
    Intensity33 INT,
```

```
    Intensity34 INT,
    Intensity35 INT,
    Intensity36 INT,
    Intensity37 INT,
    Intensity38 INT,
    Intensity39 INT,
    Intensity40 INT,
    Intensity41 INT,
    Intensity42 INT,
    Intensity43 INT,
    Intensity44 INT,
    Intensity45 INT,
    Intensity46 INT,
    Intensity47 INT,
    Intensity48 INT,
    Intensity49 INT,
    Intensity50 INT,
    Intensity51 INT,
    Intensity52 INT,
    Intensity53 INT,
    Intensity54 INT,
    Intensity55 INT,
    Intensity56 INT,
    Intensity57 INT,
    Intensity58 INT,
    Intensity59 INT
);
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteIntensitiesWide_merged.csv"
INTO TABLE minuteintensitieswide
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS

(@Id, @ActivityHour,
 @Intensity00, @Intensity01, @Intensity02, @Intensity03, @Intensity04,
@Intensity05, @Intensity06, @Intensity07, @Intensity08, @Intensity09,
 @Intensity10, @Intensity11, @Intensity12, @Intensity13, @Intensity14,
@Intensity15, @Intensity16, @Intensity17, @Intensity18, @Intensity19,
 @Intensity20, @Intensity21, @Intensity22, @Intensity23, @Intensity24,
@Intensity25, @Intensity26, @Intensity27, @Intensity28, @Intensity29,
 @Intensity30, @Intensity31, @Intensity32, @Intensity33, @Intensity34,
@Intensity35, @Intensity36, @Intensity37, @Intensity38, @Intensity39,
 @Intensity40, @Intensity41, @Intensity42, @Intensity43, @Intensity44,
@Intensity45, @Intensity46, @Intensity47, @Intensity48, @Intensity49,
 @Intensity50, @Intensity51, @Intensity52, @Intensity53, @Intensity54,
@Intensity55, @Intensity56, @Intensity57, @Intensity58, @Intensity59)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    Intensity00 = NULLIF(@Intensity00, ''),
    Intensity01 = NULLIF(@Intensity01, ''),
    Intensity02 = NULLIF(@Intensity02, ''),
    Intensity03 = NULLIF(@Intensity03, ''),
    Intensity04 = NULLIF(@Intensity04, ''),
    Intensity05 = NULLIF(@Intensity05, ''),
    Intensity06 = NULLIF(@Intensity06, ''),
```

```
    Intensity07 = NULLIF(@Intensity07, ''),
    Intensity08 = NULLIF(@Intensity08, ''),
    Intensity09 = NULLIF(@Intensity09, ''),
    Intensity10 = NULLIF(@Intensity10, ''),
    Intensity11 = NULLIF(@Intensity11, ''),
    Intensity12 = NULLIF(@Intensity12, ''),
    Intensity13 = NULLIF(@Intensity13, ''),
    Intensity14 = NULLIF(@Intensity14, ''),
    Intensity15 = NULLIF(@Intensity15, ''),
    Intensity16 = NULLIF(@Intensity16, ''),
    Intensity17 = NULLIF(@Intensity17, ''),
    Intensity18 = NULLIF(@Intensity18, ''),
    Intensity19 = NULLIF(@Intensity19, ''),
    Intensity20 = NULLIF(@Intensity20, ''),
    Intensity21 = NULLIF(@Intensity21, ''),
    Intensity22 = NULLIF(@Intensity22, ''),
    Intensity23 = NULLIF(@Intensity23, ''),
    Intensity24 = NULLIF(@Intensity24, ''),
    Intensity25 = NULLIF(@Intensity25, ''),
    Intensity26 = NULLIF(@Intensity26, ''),
    Intensity27 = NULLIF(@Intensity27, ''),
    Intensity28 = NULLIF(@Intensity28, ''),
    Intensity29 = NULLIF(@Intensity29, ''),
    Intensity30 = NULLIF(@Intensity30, ''),
    Intensity31 = NULLIF(@Intensity31, ''),
    Intensity32 = NULLIF(@Intensity32, ''),
    Intensity33 = NULLIF(@Intensity33, ''),
    Intensity34 = NULLIF(@Intensity34, ''),
    Intensity35 = NULLIF(@Intensity35, ''),
    Intensity36 = NULLIF(@Intensity36, ''),
    Intensity37 = NULLIF(@Intensity37, ''),
    Intensity38 = NULLIF(@Intensity38, ''),
    Intensity39 = NULLIF(@Intensity39, ''),
    Intensity40 = NULLIF(@Intensity40, ''),
    Intensity41 = NULLIF(@Intensity41, ''),
    Intensity42 = NULLIF(@Intensity42, ''),
    Intensity43 = NULLIF(@Intensity43, ''),
    Intensity44 = NULLIF(@Intensity44, ''),
    Intensity45 = NULLIF(@Intensity45, ''),
    Intensity46 = NULLIF(@Intensity46, ''),
    Intensity47 = NULLIF(@Intensity47, ''),
    Intensity48 = NULLIF(@Intensity48, ''),
    Intensity49 = NULLIF(@Intensity49, ''),
    Intensity50 = NULLIF(@Intensity50, ''),
    Intensity51 = NULLIF(@Intensity51, ''),
    Intensity52 = NULLIF(@Intensity52, ''),
    Intensity53 = NULLIF(@Intensity53, ''),
    Intensity54 = NULLIF(@Intensity54, ''),
    Intensity55 = NULLIF(@Intensity55, ''),
    Intensity56 = NULLIF(@Intensity56, ''),
    Intensity57 = NULLIF(@Intensity57, ''),
    Intensity58 = NULLIF(@Intensity58, ''),
    Intensity59 = NULLIF(@Intensity59, '');
```

13. minutemetsnarrow

```
CREATE TABLE minutemetsnarrow (
    Id BIGINT,
    ActivityMinute DATETIME,
```

```
    METs INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteMETsNarrow_merged.csv"
INTO TABLE minutemetsnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @METs)
SET
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    METs = NULLIF(@METs, '');
```

14. Minutesleep

```
CREATE TABLE minutesleep (
    Id BIGINT,
    Date DATETIME,
    Value INT,
    LogId BIGINT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteSleep_merged.csv"
INTO TABLE minutesleep
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @Date, @Value, @LogId)
SET
    Id = NULLIF(@Id, ''),
    Date = STR_TO_DATE(@Date, '%m/%d/%Y %r'),
    Value = NULLIF(@Value, ''),
    LogId = NULLIF(@LogId, '');
```

15. Minutestepsnarrow

```
CREATE TABLE minutestepsnarrow (
    Id BIGINT,
    ActivityMinute DATETIME,
    Steps INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteStepsNarrow_merged.csv"
INTO TABLE minutestepsnarrow
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityMinute, @Steps)
SET
```

```
    Id = NULLIF(@Id, ''),
    ActivityMinute = STR_TO_DATE(@ActivityMinute, '%m/%d/%Y %r'),
    Steps = NULLIF(@Steps, '');
```

16. minutestepswide

```
CREATE TABLE minutestepswide (
    Id BIGINT,
    ActivityHour DATETIME,
    Steps00 INT,
    Steps01 INT,
    Steps02 INT,
    Steps03 INT,
    Steps04 INT,
    Steps05 INT,
    Steps06 INT,
    Steps07 INT,
    Steps08 INT,
    Steps09 INT,
    Steps10 INT,
    Steps11 INT,
    Steps12 INT,
    Steps13 INT,
    Steps14 INT,
    Steps15 INT,
    Steps16 INT,
    Steps17 INT,
    Steps18 INT,
    Steps19 INT,
    Steps20 INT,
    Steps21 INT,
    Steps22 INT,
    Steps23 INT,
    Steps24 INT,
    Steps25 INT,
    Steps26 INT,
    Steps27 INT,
    Steps28 INT,
    Steps29 INT,
    Steps30 INT,
    Steps31 INT,
    Steps32 INT,
    Steps33 INT,
    Steps34 INT,
    Steps35 INT,
    Steps36 INT,
    Steps37 INT,
    Steps38 INT,
    Steps39 INT,
    Steps40 INT,
    Steps41 INT,
    Steps42 INT,
    Steps43 INT,
    Steps44 INT,
    Steps45 INT,
    Steps46 INT,
    Steps47 INT,
    Steps48 INT,
```

```
    Steps49 INT,
    Steps50 INT,
    Steps51 INT,
    Steps52 INT,
    Steps53 INT,
    Steps54 INT,
    Steps55 INT,
    Steps56 INT,
    Steps57 INT,
    Steps58 INT,
    Steps59 INT
);
LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/minuteStepsWide_merged.csv"
INTO TABLE minutestepswide
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @ActivityHour,
 @Steps00, @Steps01, @Steps02, @Steps03, @Steps04, @Steps05, @Steps06,
@Steps07, @Steps08, @Steps09,
 @Steps10, @Steps11, @Steps12, @Steps13, @Steps14, @Steps15, @Steps16,
@Steps17, @Steps18, @Steps19,
 @Steps20, @Steps21, @Steps22, @Steps23, @Steps24, @Steps25, @Steps26,
@Steps27, @Steps28, @Steps29,
 @Steps30, @Steps31, @Steps32, @Steps33, @Steps34, @Steps35, @Steps36,
@Steps37, @Steps38, @Steps39,
 @Steps40, @Steps41, @Steps42, @Steps43, @Steps44, @Steps45, @Steps46,
@Steps47, @Steps48, @Steps49,
 @Steps50, @Steps51, @Steps52, @Steps53, @Steps54, @Steps55, @Steps56,
@Steps57, @Steps58, @Steps59)
SET
    Id = NULLIF(@Id, ''),
    ActivityHour = STR_TO_DATE(@ActivityHour, '%m/%d/%Y %r'),
    Steps00 = NULLIF(@Steps00, ''),
    Steps01 = NULLIF(@Steps01, ''),
    Steps02 = NULLIF(@Steps02, ''),
    Steps03 = NULLIF(@Steps03, ''),
    Steps04 = NULLIF(@Steps04, ''),
    Steps05 = NULLIF(@Steps05, ''),
    Steps06 = NULLIF(@Steps06, ''),
    Steps07 = NULLIF(@Steps07, ''),
    Steps08 = NULLIF(@Steps08, ''),
    Steps09 = NULLIF(@Steps09, ''),
    Steps10 = NULLIF(@Steps10, ''),
    Steps11 = NULLIF(@Steps11, ''),
    Steps12 = NULLIF(@Steps12, ''),
    Steps13 = NULLIF(@Steps13, ''),
    Steps14 = NULLIF(@Steps14, ''),
    Steps15 = NULLIF(@Steps15, ''),
    Steps16 = NULLIF(@Steps16, ''),
    Steps17 = NULLIF(@Steps17, ''),
    Steps18 = NULLIF(@Steps18, ''),
    Steps19 = NULLIF(@Steps19, ''),
    Steps20 = NULLIF(@Steps20, ''),
    Steps21 = NULLIF(@Steps21, ''),
    Steps22 = NULLIF(@Steps22, ''),
```

```
    Steps23 = NULLIF(@Steps23, ''),
    Steps24 = NULLIF(@Steps24, ''),
    Steps25 = NULLIF(@Steps25, ''),
    Steps26 = NULLIF(@Steps26, ''),
    Steps27 = NULLIF(@Steps27, ''),
    Steps28 = NULLIF(@Steps28, ''),
    Steps29 = NULLIF(@Steps29, ''),
    Steps30 = NULLIF(@Steps30, ''),
    Steps31 = NULLIF(@Steps31, ''),
    Steps32 = NULLIF(@Steps32, ''),
    Steps33 = NULLIF(@Steps33, ''),
    Steps34 = NULLIF(@Steps34, ''),
    Steps35 = NULLIF(@Steps35, ''),
    Steps36 = NULLIF(@Steps36, ''),
    Steps37 = NULLIF(@Steps37, ''),
    Steps38 = NULLIF(@Steps38, ''),
    Steps39 = NULLIF(@Steps39, ''),
    Steps40 = NULLIF(@Steps40, ''),
    Steps41 = NULLIF(@Steps41, ''),
    Steps42 = NULLIF(@Steps42, ''),
    Steps43 = NULLIF(@Steps43, ''),
    Steps44 = NULLIF(@Steps44, ''),
    Steps45 = NULLIF(@Steps45, ''),
    Steps46 = NULLIF(@Steps46, ''),
    Steps47 = NULLIF(@Steps47, ''),
    Steps48 = NULLIF(@Steps48, ''),
    Steps49 = NULLIF(@Steps49, ''),
    Steps50 = NULLIF(@Steps50, ''),
    Steps51 = NULLIF(@Steps51, ''),
    Steps52 = NULLIF(@Steps52, ''),
    Steps53 = NULLIF(@Steps53, ''),
    Steps54 = NULLIF(@Steps54, ''),
    Steps55 = NULLIF(@Steps55, ''),
    Steps56 = NULLIF(@Steps56, ''),
    Steps57 = NULLIF(@Steps57, ''),
    Steps58 = NULLIF(@Steps58, ''),
    Steps59 = NULLIF(@Steps59, '');
```

17. sleepday

```
CREATE TABLE sleepday (
    Id BIGINT,
    SleepDay DATETIME,
    TotalSleepRecords INT,
    TotalMinutesAsleep INT,
    TotalTimeInBed INT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/sleepDay_merged.csv"
INTO TABLE sleepday
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @SleepDay, @TotalSleepRecords, @TotalMinutesAsleep,
@TotalTimeInBed)
SET
```

```
    Id = NULLIF(@Id, ''),
    SleepDay = STR_TO_DATE(@SleepDay, '%m/%d/%Y %r'),
    TotalSleepRecords = NULLIF(@TotalSleepRecords, ''),
    TotalMinutesAsleep = NULLIF(@TotalMinutesAsleep, ''),
    TotalTimeInBed = NULLIF(@TotalTimeInBed, '');
```

18. weightloginfo

```
CREATE TABLE weightloginfo (
    Id BIGINT,
    Date DATETIME,
    WeightKg DOUBLE,
    WeightPounds DOUBLE,
    Fat BIGINT,
    BMI DOUBLE,
    IsManualReport BOOLEAN,
    LogId BIGINT
);

LOAD DATA LOCAL INFILE "C:/Users/dhara/Documents/Interns/Labmentix/6.
Starva Fitness/Data Files/weightLogInfo_merged.csv"
INTO TABLE weightloginfo
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS
(@Id, @Date, @WeightKg, @WeightPounds, @Fat, @BMI, @IsManualReport,
@LogId)
SET
    Id = NULLIF(@Id, ''),
    Date = STR_TO_DATE(@Date, '%m/%d/%Y %r'),
    WeightKg = NULLIF(@WeightKg, ''),
      WeightPounds = NULLIF(@WeightPounds, ''),
    Fat = NULLIF(@Fat, ''),
    BMI = NULLIF(@BMI, ''),
    IsManualReport = IF(@IsManualReport = 'TRUE', TRUE, FALSE),
    LogId = NULLIF(@LogId, '');
```

MERGE DAILY DATA INTO ONE TABLE

```
CREATE TABLE daily_merged_data AS
SELECT
     -- dailyactivity
    da.Id,
    da.ActivityDate,
    da.TotalSteps,
    da.TotalDistance,
    da.TrackerDistance,
    da.LoggedActivitiesDistance,
    da.VeryActiveDistance,
    da.ModeratelyActiveDistance,
    da.LightActiveDistance,
    da.SedentaryActiveDistance,
    da.VeryActiveMinutes,
    da.FairlyActiveMinutes,
    da.LightlyActiveMinutes,
    da.SedentaryMinutes,
    da.Calories AS activity_calories,
```

```
    -- Daily Calories
    dc.Calories AS daily_calories,

    -- Daily Intensities
    di.SedentaryMinutes AS di_SedentaryMinutes,
    di.LightlyActiveMinutes AS di_LightlyActiveMinutes,
    di.FairlyActiveMinutes AS di_FairlyActiveMinutes,
    di.VeryActiveMinutes AS di_VeryActiveMinutes,
    di.SedentaryActiveDistance AS di_SedentaryActiveDistance,
    di.LightActiveDistance AS di_LightActiveDistance,
    di.ModeratelyActiveDistance AS di_ModeratelyActiveDistance,
    di.VeryActiveDistance AS di_VeryActiveDistance,

    -- Daily Steps
    ds.StepTotal AS ds_step_total,

    -- Heart Rate (daily average)
    hr.avg_heart_rate,

    -- Sleep Day
    sl.TotalSleepRecords,
    sl.TotalMinutesAsleep,
    sl.TotalTimeInBed,

    -- weight log info
    wl.WeightKg,
    wl.WeightPounds,
    wl.Fat,
    wl.BMI,
    wl.LogId

FROM dailyactivity AS da

-- JOIN with daily_calories
LEFT JOIN dailycalories AS dc
  ON da.Id = dc.Id AND da.ActivityDate = dc.ActivityDay

-- JOIN with daily_intensities
LEFT JOIN dailyintensities as di
  ON da.Id = di.Id AND da.ActivityDate = di.ActivityDay

-- JOIN with daily_steps
LEFT JOIN dailysteps as ds
  ON da.Id = ds.Id AND da.ActivityDate = ds.ActivityDay

-- JOIN with average daily heart rate
LEFT JOIN (
    SELECT
        Id,
        DATE(Time) AS HR_Date,
        ROUND(AVG(Value), 2) AS avg_heart_rate
    FROM heartrateseconds
    GROUP BY Id, HR_Date
) as hr
  ON da.Id = hr.Id AND da.ActivityDate = hr.HR_Date

-- JOIN with sleep data
```

```
LEFT JOIN sleepday as sl
  ON da.Id = sl.Id AND da.ActivityDate = date(sl.SleepDay)

-- JOIN with weight data
LEFT JOIN (
    SELECT
        Id,
        DATE(Date) AS WeightDate,
        WeightKg,
        WeightPounds,
        Fat,
        BMI,
        LogId
    FROM weightloginfo
) AS wl
  ON da.Id = wl.Id AND da.ActivityDate = wl.WeightDate;

-- MERGE TIME BASED DATA INTO ANOTHER TABLE

CREATE TABLE timebased_merged_data AS
SELECT
    -- hourly calories
    hc.Id,
    hc.ActivityHour AS TimeStamp,
      hc.Calories AS hourly_calories,

    -- hourly intensities
    hi.TotalIntensity AS hourly_total_intensity,
    hi.AverageIntensity AS hourly_avg_intensity,

    -- hourly steps
    hs.StepTotal AS hourly_steps,

    -- minute calories narrow
    mcn.Calories AS minute_calories_narrow,

    -- minute calories wide
     mcw.Calories00 AS calories_wide00,
       mcw.Calories01 AS calories_wide01,
       mcw.Calories02 AS calories_wide02,
       mcw.Calories03 AS calories_wide03,
       mcw.Calories04 AS calories_wide04,
     mcw.Calories05 AS calories_wide05,
     mcw.Calories06 AS calories_wide06,
     mcw.Calories07 AS calories_wide07,
     mcw.Calories08 AS calories_wide08,
     mcw.Calories09 AS calories_wide09,
     mcw.Calories10 AS calories_wide10,
     mcw.Calories11 AS calories_wide11,
     mcw.Calories12 AS calories_wide12,
     mcw.Calories13 AS calories_wide13,
     mcw.Calories14 AS calories_wide14,
     mcw.Calories15 AS calories_wide15,
     mcw.Calories16 AS calories_wide16,
     mcw.Calories17 AS calories_wide17,
     mcw.Calories18 AS calories_wide18,
     mcw.Calories19 AS calories_wide19,
     mcw.Calories20 AS calories_wide20,
```

```
 mcw.Calories21 AS calories_wide21,
 mcw.Calories22 AS calories_wide22,
 mcw.Calories23 AS calories_wide23,
 mcw.Calories24 AS calories_wide24,
 mcw.Calories25 AS calories_wide25,
 mcw.Calories26 AS calories_wide26,
   mcw.Calories27 AS calories_wide27,
   mcw.Calories28 AS calories_wide28,
 mcw.Calories29 AS calories_wide29,
   mcw.Calories30 AS calories_wide30,
 mcw.Calories31 AS calories_wide31,
 mcw.Calories32 AS calories_wide32,
 mcw.Calories33 AS calories_wide33,
 mcw.Calories34 AS calories_wide34,
 mcw.Calories35 AS calories_wide35,
   mcw.Calories36 AS calories_wide36,
 mcw.Calories37 AS calories_wide37,
 mcw.Calories38 AS calories_wide38,
 mcw.Calories39 AS calories_wide39,
   mcw.Calories40 AS calories_wide40,
 mcw.Calories41 AS calories_wide41,
 mcw.Calories42 AS calories_wide42,
 mcw.Calories43 AS calories_wide43,
 mcw.Calories44 AS calories_wide44,
 mcw.Calories45 AS calories_wide45,
 mcw.Calories46 AS calories_wide46,
 mcw.Calories47 AS calories_wide47,
 mcw.Calories48 AS calories_wide48,
 mcw.Calories49 AS calories_wide49,
 mcw.Calories50 AS calories_wide50,
 mcw.Calories51 AS calories_wide51,
 mcw.Calories52 AS calories_wide52,
 mcw.Calories53 AS calories_wide53,
 mcw.Calories54 AS calories_wide54,
 mcw.Calories55 AS calories_wide55,
 mcw.Calories56 AS calories_wide56,
 mcw.Calories57 AS calories_wide57,
 mcw.Calories58 AS calories_wide58,
 mcw.Calories59 AS calories_wide59,

 -- minute intensities narrow
 minin.Intensity AS minute_intensity_narrow,

-- minute intensities wide
 miniw.Intensity00 AS intensity_wide00,
 miniw.Intensity01 AS intensity_wide01,
 miniw.Intensity02 AS intensity_wide02,
   miniw.Intensity03 AS intensity_wide03,
 miniw.Intensity04 AS intensity_wide04,
 miniw.Intensity05 AS intensity_wide05,
 miniw.Intensity06 AS intensity_wide06,
 miniw.Intensity07 AS intensity_wide07,
 miniw.Intensity08 AS intensity_wide08,
 miniw.Intensity09 AS intensity_wide09,
 miniw.Intensity10 AS intensity_wide10,
 miniw.Intensity11 AS intensity_wide11,
 miniw.Intensity12 AS intensity_wide12,
 miniw.Intensity13 AS intensity_wide13,
```

```
 miniw.Intensity14 AS intensity_wide14,
 miniw.Intensity15 AS intensity_wide15,
   miniw.Intensity16 AS intensity_wide16,
 miniw.Intensity17 AS intensity_wide17,
 miniw.Intensity18 AS intensity_wide18,
 miniw.Intensity19 AS intensity_wide19,
 miniw.Intensity20 AS intensity_wide20,
 miniw.Intensity21 AS intensity_wide21,
 miniw.Intensity22 AS intensity_wide22,
 miniw.Intensity23 AS intensity_wide23,
 miniw.Intensity24 AS intensity_wide24,
 miniw.Intensity25 AS intensity_wide25,
 miniw.Intensity26 AS intensity_wide26,
 miniw.Intensity27 AS intensity_wide27,
   miniw.Intensity28 AS intensity_wide28,
 miniw.Intensity29 AS intensity_wide29,
 miniw.Intensity30 AS intensity_wide30,
 miniw.Intensity31 AS intensity_wide31,
 miniw.Intensity32 AS intensity_wide32,
 miniw.Intensity33 AS intensity_wide33,
 miniw.Intensity34 AS intensity_wide34,
 miniw.Intensity35 AS intensity_wide35,
 miniw.Intensity36 AS intensity_wide36,
 miniw.Intensity37 AS intensity_wide37,
   miniw.Intensity38 AS intensity_wide38,
 miniw.Intensity39 AS intensity_wide39,
 miniw.Intensity40 AS intensity_wide40,
 miniw.Intensity41 AS intensity_wide41,
 miniw.Intensity42 AS intensity_wide42,
 miniw.Intensity43 AS intensity_wide43,
 miniw.Intensity44 AS intensity_wide44,
 miniw.Intensity45 AS intensity_wide45,
 miniw.Intensity46 AS intensity_wide46,
 miniw.Intensity47 AS intensity_wide47,
 miniw.Intensity48 AS intensity_wide48,
 miniw.Intensity49 AS intensity_wide49,
 miniw.Intensity50 AS intensity_wide50,
 miniw.Intensity51 AS intensity_wide51,
 miniw.Intensity52 AS intensity_wide52,
 miniw.Intensity53 AS intensity_wide53,
 miniw.Intensity54 AS intensity_wide54,
 miniw.Intensity55 AS intensity_wide55,
 miniw.Intensity56 AS intensity_wide56,
 miniw.Intensity57 AS intensity_wide57,
 miniw.Intensity58 AS intensity_wide58,
 miniw.Intensity59 AS intensity_wide59,

 -- minute met snarrow
 mm.METs AS minute_mets,

-- minutesleep
 ms.Value AS minute_sleep_value,

 -- minute steps narrow
  msn.Steps AS minute_steps_narrow,

  -- minute steps wide
 msw.Steps00 AS minute_steps_wide00,
```

```
msw.Steps01 AS minute_steps_wide01,
msw.Steps02 AS minute_steps_wide02,
msw.Steps03 AS minute_steps_wide03,
  msw.Steps04 AS minute_steps_wide04,
msw.Steps05 AS minute_steps_wide05,
msw.Steps06 AS minute_steps_wide06,
msw.Steps07 AS minute_steps_wide07,
msw.Steps08 AS minute_steps_wide08,
msw.Steps09 AS minute_steps_wide09,
msw.Steps10 AS minute_steps_wide10,
msw.Steps11 AS minute_steps_wide11,
msw.Steps12 AS minute_steps_wide12,
msw.Steps13 AS minute_steps_wide13,
msw.Steps14 AS minute_steps_wide14,
msw.Steps15 AS minute_steps_wide15,
msw.Steps16 AS minute_steps_wide16,
msw.Steps17 AS minute_steps_wide17,
msw.Steps18 AS minute_steps_wide18,
msw.Steps19 AS minute_steps_wide19,
msw.Steps20 AS minute_steps_wide20,
msw.Steps21 AS minute_steps_wide21,
msw.Steps22 AS minute_steps_wide22,
msw.Steps23 AS minute_steps_wide23,
msw.Steps24 AS minute_steps_wide24,
msw.Steps25 AS minute_steps_wide25,
msw.Steps26 AS minute_steps_wide26,
msw.Steps27 AS minute_steps_wide27,
msw.Steps28 AS minute_steps_wide28,
msw.Steps29 AS minute_steps_wide29,
msw.Steps30 AS minute_steps_wide30,
msw.Steps31 AS minute_steps_wide31,
msw.Steps32 AS minute_steps_wide32,
msw.Steps33 AS minute_steps_wide33,
msw.Steps34 AS minute_steps_wide34,
msw.Steps35 AS minute_steps_wide35,
msw.Steps36 AS minute_steps_wide36,
msw.Steps37 AS minute_steps_wide37,
msw.Steps38 AS minute_steps_wide38,
msw.Steps39 AS minute_steps_wide39,
msw.Steps40 AS minute_steps_wide40,
msw.Steps41 AS minute_steps_wide41,
msw.Steps42 AS minute_steps_wide42,
msw.Steps43 AS minute_steps_wide43,
msw.Steps44 AS minute_steps_wide44,
msw.Steps45 AS minute_steps_wide45,
msw.Steps46 AS minute_steps_wide46,
msw.Steps47 AS minute_steps_wide47,
msw.Steps48 AS minute_steps_wide48,
msw.Steps49 AS minute_steps_wide49,
msw.Steps50 AS minute_steps_wide50,
  msw.Steps51 AS minute_steps_wide51,
msw.Steps52 AS minute_steps_wide52,
msw.Steps53 AS minute_steps_wide53,
msw.Steps54 AS minute_steps_wide54,
msw.Steps56 AS minute_steps_wide56,
msw.Steps57 AS minute_steps_wide57,
msw.Steps58 AS minute_steps_wide58,
msw.Steps59 AS minute_steps_wide59
```

```
FROM hourlycalories as hc

-- JOIN hourly intensities
LEFT JOIN hourlyintensities as hi
    ON hc.Id = hi.Id AND hc.ActivityHour = hi.ActivityHour

-- JOIN hourly steps
LEFT JOIN hourlysteps as hs
    ON hc.Id = hs.Id AND hc.ActivityHour = hs.ActivityHour

-- JOIN minuteCaloriesNarrow
LEFT JOIN minutecaloriesnarrow as mcn
    ON hc.Id = mcn.Id AND hc.ActivityHour = mcn.ActivityMinute

-- JOIN minuteCaloriesWide
LEFT JOIN minutecalorieswide as mcw
    ON hc.Id = mcw.Id AND hc.ActivityHour = mcw.ActivityMinute

-- JOIN minuteIntensitiesNarrow
LEFT JOIN minuteintensitiesnarrow as minin
    ON hc.Id = minin.Id AND hc.ActivityHour = minin.ActivityMinute

-- JOIN minuteIntensitiesWide
LEFT JOIN minuteintensitieswide as miniw
    ON hc.Id = miniw.Id AND hc.ActivityHour = miniw.ActivityHour

-- JOIN minuteMETsNarrow
LEFT JOIN minutemetsnarrow as mm
    ON hc.Id = mm.Id AND hc.ActivityHour = mm.ActivityMinute

-- JOIN minuteSleep
LEFT JOIN minutesleep as ms
    ON hc.Id = ms.Id AND hc.ActivityHour = ms.date

-- JOIN minuteStepsNarrow
LEFT JOIN minutestepsnarrow as msn
    ON hc.Id = msn.Id AND hc.ActivityHour = msn.ActivityMinute

-- JOIN minuteStepsWide
LEFT JOIN minutestepswide as msw
    ON hc.Id = msw.Id AND hc.ActivityHour = msw.ActivityHour;


-- MERGE BOTH THE TABLES INTO SINGLE TABLE

CREATE TABLE StravaFullMergedData AS
SELECT
    -- From daily_merged_data
    dmd.Id,
    dmd.ActivityDate,
    dmd.TotalSteps,
    dmd.TotalDistance,
    dmd.TrackerDistance,
    dmd.LoggedActivitiesDistance,
    dmd.VeryActiveDistance,
    dmd.ModeratelyActiveDistance,
```

```
    dmd.LightActiveDistance,
    dmd.SedentaryActiveDistance,
    dmd.VeryActiveMinutes,
    dmd.FairlyActiveMinutes,
    dmd.LightlyActiveMinutes,
    dmd.SedentaryMinutes,
    dmd.activity_calories,
      dmd.daily_calories,
      dmd.di_SedentaryMinutes,
    dmd.di_LightlyActiveMinutes,
    dmd.di_FairlyActiveMinutes,
    dmd.di_VeryActiveMinutes,
      dmd.di_SedentaryActiveDistance,
    dmd.di_LightActiveDistance,
    dmd.di_ModeratelyActiveDistance,
    dmd.di_VeryActiveDistance,
    dmd.ds_step_total,
    dmd.avg_heart_rate,
    dmd.TotalSleepRecords,
    dmd.TotalMinutesAsleep,
    dmd.TotalTimeInBed,
    dmd.WeightKg,
    dmd.Fat,
    dmd.BMI,

    -- From timebased_merged_data
    tmd.TimeStamp,
    tmd.hourly_calories,
    tmd.hourly_total_intensity,
    tmd.hourly_avg_intensity,
    tmd.hourly_steps,
    tmd.minute_calories_narrow,
    tmd.minute_intensity_narrow,
    tmd.minute_mets,
    tmd.minute_sleep_value,
    tmd.minute_steps_narrow,

    -- Calories Wide Columns (00 to 59)
    tmd.calories_wide00, tmd.calories_wide01, tmd.calories_wide02,
tmd.calories_wide03, tmd.calories_wide04,
    tmd.calories_wide05, tmd.calories_wide06, tmd.calories_wide07,
tmd.calories_wide08, tmd.calories_wide09,
    tmd.calories_wide10, tmd.calories_wide11, tmd.calories_wide12,
tmd.calories_wide13, tmd.calories_wide14,
    tmd.calories_wide15, tmd.calories_wide16, tmd.calories_wide17,
tmd.calories_wide18, tmd.calories_wide19,
    tmd.calories_wide20, tmd.calories_wide21, tmd.calories_wide22,
tmd.calories_wide23, tmd.calories_wide24,
    tmd.calories_wide25, tmd.calories_wide26, tmd.calories_wide27,
tmd.calories_wide28, tmd.calories_wide29,
    tmd.calories_wide30, tmd.calories_wide31, tmd.calories_wide32,
tmd.calories_wide33, tmd.calories_wide34,
    tmd.calories_wide35, tmd.calories_wide36, tmd.calories_wide37,
tmd.calories_wide38, tmd.calories_wide39,
    tmd.calories_wide40, tmd.calories_wide41, tmd.calories_wide42,
tmd.calories_wide43, tmd.calories_wide44,
    tmd.calories_wide45, tmd.calories_wide46, tmd.calories_wide47,
tmd.calories_wide48, tmd.calories_wide49,
```

```
    tmd.calories_wide50, tmd.calories_wide51, tmd.calories_wide52,
tmd.calories_wide53, tmd.calories_wide54,
    tmd.calories_wide55, tmd.calories_wide56, tmd.calories_wide57,
tmd.calories_wide58, tmd.calories_wide59,

    -- Intensity Wide Columns (00 to 59)
    tmd.intensity_wide00, tmd.intensity_wide01, tmd.intensity_wide02,
tmd.intensity_wide03, tmd.intensity_wide04,
    tmd.intensity_wide05, tmd.intensity_wide06, tmd.intensity_wide07,
tmd.intensity_wide08, tmd.intensity_wide09,
    tmd.intensity_wide10, tmd.intensity_wide11, tmd.intensity_wide12,
tmd.intensity_wide13, tmd.intensity_wide14,
    tmd.intensity_wide15, tmd.intensity_wide16, tmd.intensity_wide17,
tmd.intensity_wide18, tmd.intensity_wide19,
    tmd.intensity_wide20, tmd.intensity_wide21, tmd.intensity_wide22,
tmd.intensity_wide23, tmd.intensity_wide24,
    tmd.intensity_wide25, tmd.intensity_wide26, tmd.intensity_wide27,
tmd.intensity_wide28, tmd.intensity_wide29,
    tmd.intensity_wide30, tmd.intensity_wide31, tmd.intensity_wide32,
tmd.intensity_wide33, tmd.intensity_wide34,
    tmd.intensity_wide35, tmd.intensity_wide36, tmd.intensity_wide37,
tmd.intensity_wide38, tmd.intensity_wide39,
    tmd.intensity_wide40, tmd.intensity_wide41, tmd.intensity_wide42,
tmd.intensity_wide43, tmd.intensity_wide44,
    tmd.intensity_wide45, tmd.intensity_wide46, tmd.intensity_wide47,
tmd.intensity_wide48, tmd.intensity_wide49,
    tmd.intensity_wide50, tmd.intensity_wide51, tmd.intensity_wide52,
tmd.intensity_wide53, tmd.intensity_wide54,
    tmd.intensity_wide55, tmd.intensity_wide56, tmd.intensity_wide57,
tmd.intensity_wide58, tmd.intensity_wide59,

    -- Steps Wide Columns (00 to 59)
    tmd.minute_steps_wide00, tmd.minute_steps_wide01,
tmd.minute_steps_wide02, tmd.minute_steps_wide03,
tmd.minute_steps_wide04,
    tmd.minute_steps_wide05, tmd.minute_steps_wide06,
tmd.minute_steps_wide07, tmd.minute_steps_wide08,
tmd.minute_steps_wide09,
    tmd.minute_steps_wide10, tmd.minute_steps_wide11,
tmd.minute_steps_wide12, tmd.minute_steps_wide13,
tmd.minute_steps_wide14,
    tmd.minute_steps_wide15, tmd.minute_steps_wide16,
tmd.minute_steps_wide17, tmd.minute_steps_wide18,
tmd.minute_steps_wide19,
    tmd.minute_steps_wide20, tmd.minute_steps_wide21,
tmd.minute_steps_wide22, tmd.minute_steps_wide23,
tmd.minute_steps_wide24,
    tmd.minute_steps_wide25, tmd.minute_steps_wide26,
tmd.minute_steps_wide27, tmd.minute_steps_wide28,
tmd.minute_steps_wide29,
    tmd.minute_steps_wide30, tmd.minute_steps_wide31,
tmd.minute_steps_wide32, tmd.minute_steps_wide33,
tmd.minute_steps_wide34,
    tmd.minute_steps_wide35, tmd.minute_steps_wide36,
tmd.minute_steps_wide37, tmd.minute_steps_wide38,
tmd.minute_steps_wide39,
```

```
    tmd.minute_steps_wide40, tmd.minute_steps_wide41,
tmd.minute_steps_wide42, tmd.minute_steps_wide43,
tmd.minute_steps_wide44,
    tmd.minute_steps_wide45, tmd.minute_steps_wide46,
tmd.minute_steps_wide47, tmd.minute_steps_wide48,
tmd.minute_steps_wide49,
    tmd.minute_steps_wide50, tmd.minute_steps_wide51,
tmd.minute_steps_wide52, tmd.minute_steps_wide53,
tmd.minute_steps_wide54,
    tmd.minute_steps_wide56, tmd.minute_steps_wide57,
tmd.minute_steps_wide58, tmd.minute_steps_wide59

FROM
    daily_merged_data AS dmd
LEFT JOIN
    timebased_merged_data AS tmd
    ON dmd.Id = tmd.Id AND dmd.ActivityDate = DATE(tmd.TimeStamp);
```

PREPROCESSING STEPS

```
-- check datatype
 DESCRIBE stravafullmergeddata;

-- Handle missing values

SET SQL_SAFE_UPDATES = 0;

-- avg_heart_rate
UPDATE stravafullmergeddata
SET avg_heart_rate = (
    SELECT ROUND(AVG(avg_heart_rate), 1)
    FROM (SELECT avg_heart_rate FROM stravafullmergeddata WHERE
avg_heart_rate IS NOT NULL) AS t
)
WHERE avg_heart_rate IS NULL;

-- TotalSleepRecords
UPDATE stravafullmergeddata
SET TotalSleepRecords = (
    SELECT ROUND(AVG(TotalSleepRecords), 1)
    FROM (SELECT TotalSleepRecords FROM stravafullmergeddata WHERE
TotalSleepRecords IS NOT NULL) AS t
)
WHERE TotalSleepRecords IS NULL;

-- TotalMinutesAsleep
UPDATE stravafullmergeddata
SET TotalMinutesAsleep = (
    SELECT ROUND(AVG(TotalMinutesAsleep), 1)
    FROM (SELECT TotalMinutesAsleep FROM stravafullmergeddata WHERE
TotalMinutesAsleep IS NOT NULL) AS t
)
WHERE TotalMinutesAsleep IS NULL;

-- TotalTimeInBed
UPDATE stravafullmergeddata
SET TotalTimeInBed = (
```

```
    SELECT ROUND(AVG(TotalTimeInBed), 1)
    FROM (SELECT TotalTimeInBed FROM stravafullmergeddata WHERE
TotalTimeInBed IS NOT NULL) AS t
)
WHERE TotalTimeInBed IS NULL;

-- WeightKg
UPDATE stravafullmergeddata
SET WeightKg = (
    SELECT ROUND(AVG(WeightKg), 1)
    FROM (SELECT WeightKg FROM stravafullmergeddata WHERE WeightKg IS
NOT NULL) AS t
)
WHERE WeightKg IS NULL;

-- Fat
UPDATE stravafullmergeddata
SET Fat = (
    SELECT ROUND(AVG(Fat), 1)
    FROM (SELECT Fat FROM stravafullmergeddata WHERE Fat IS NOT NULL)
AS t
)
WHERE Fat IS NULL AND Id IS NOT NULL;

-- BMI
UPDATE stravafullmergeddata
SET BMI = (
    SELECT ROUND(AVG(BMI), 2)
    FROM (SELECT BMI FROM stravafullmergeddata WHERE BMI IS NOT NULL)
AS t
)
WHERE BMI IS NULL;

-- minute_sleep_value
UPDATE stravafullmergeddata
SET minute_sleep_value = (
    SELECT ROUND(AVG(minute_sleep_value))
    FROM (SELECT minute_sleep_value FROM stravafullmergeddata WHERE
minute_sleep_value IS NOT NULL) AS t
)
WHERE minute_sleep_value IS NULL AND Id IS NOT NULL;

-- drop missing values
DELETE FROM StravaFullMergedData
WHERE calories_wide00 IS NULL;

# Data Validation
SELECT
    MIN(avg_heart_rate) AS min_hr,
    MAX(avg_heart_rate) AS max_hr,
    MIN(BMI) AS min_bmi,
    MAX(BMI) AS max_bmi
FROM starva.stravafullmergeddata;

-- check for outliners
-- BMI outside normal range
SELECT * FROM stravafullmergeddata
WHERE BMI < 10 OR BMI > 50;
```

```
-- Heart rate outliers
SELECT * FROM stravafullmergeddata
WHERE avg_heart_rate < 30 OR avg_heart_rate > 220;
```

-- Check for Duplicates

```
SELECT Id, ActivityDate, COUNT(*) AS occurrences
FROM stravafullmergeddata
GROUP BY Id, ActivityDate
HAVING COUNT(*) > 1;


SET SQL_SAFE_UPDATES = 0;
DELETE t1
FROM stravafullmergeddata t1
JOIN stravafullmergeddata t2
ON t1.Id = t2.Id
   AND t1.ActivityDate = t2.ActivityDate
   AND t1.temp_row_id > t2.temp_row_id;
ALTER TABLE stravafullmergeddata DROP COLUMN temp_row_id;
```

-- Delete Duplicates

```
ALTER TABLE stravafullmergeddata
ADD COLUMN temp_row_id INT NOT NULL AUTO_INCREMENT PRIMARY KEY;


-- ____________________________________________________________ -
-
```

BUSINESS INSIGHTS

```
-- 1. Daily Activity Table
SELECT ActivityDate FROM dailyactivity;

-- Max Distance By Max Active Distance
SELECT id, max(TotalDistance) AS TopDistance,
max(VeryActiveDistance) AS TopActiveDistance
FROM DailyActivity
GROUP BY id
ORDER BY TopDistance DESC;

-- Top Active Minutes By Calories
SELECT id, max(VeryActiveDistance) AS TopActiveMinutes,
max(TrackerDistance) AS MaxTrackerDistance, Calories
FROM DailyActivity
GROUP BY id, Calories
ORDER BY MaxTrackerDistance DESC, Calories DESC LIMIT 5;

-- Activity Date by Topsteps, TopDistance And Calories
SELECT id, ActivityDate, max(TotalSteps) AS TopSteps,
max(TotalDistance) AS TopDistance,
Calories
FROM DailyActivity
GROUP BY id, ActivityDate, Calories
ORDER BY TopSteps DESC, TopDistance DESC, Calories DESC
LIMIT 10;

-- 2. Daily Calories Table
SELECT * FROM dailycalories;
```

```
-- AvgCalories
SELECT ActivityDay, avg(Calories) AS AvgCalories
FROM dailycalories
GROUP BY ActivityDay;

-- Inner join Daily Activity by total steps
SELECT a.id, a.ActivityDate, a.TotalSteps, c.Calories
FROM dailyactivity a INNER JOIN dailycalories c ON a.id = c.id;

-- Daily Activity Date by Daily Calories (using Window Function)
SELECT ActivityDate, TotalDistance AS MaxDistance, Calories
FROM (
    SELECT a.ActivityDate, a.TotalDistance, c.Calories,
    row_number() OVER (PARTITION BY a.ActivityDate ORDER BY
a.TotalDistance DESC) AS Activity
    FROM dailyactivity a
    INNER JOIN dailycalories c ON a.id = c.id
) sub
ORDER BY MaxDistance DESC
LIMIT 10;

-- Checking for Min & Max values
SELECT
min(TotalDistance), max(TotalDistance), min(TotalSteps),
max(TotalSteps),
min(LoggedActivitiesDistance), max(LoggedActivitiesDistance),
min(SedentaryActiveDistance),
max(SedentaryActiveDistance),
min(SedentaryMinutes), max(SedentaryMinutes),
min(Calories), max(Calories)
FROM dailyactivity;

-- Checking SedentaryMinutes by Daily Patterns
SELECT id, SedentaryMinutes, count(*) AS 'Sedentary Days (=> 1440
mins)'
FROM dailyactivity
WHERE SedentaryMinutes >= 1440
GROUP BY id, SedentaryMinutes;

-- 3. Sleep Day
SELECT * FROM sleepday;

-- Avg Sleep Minutes By Id
SELECT id, avg(TotalMinutesAsleep) AS AvgSleepMinutes
FROM sleepday
GROUP BY id;

-- Sleepday Records more than 2
SELECT id, TotalSleepRecords, count(*) AS MoreSleepRecords
FROM sleepday
WHERE TotalSleepRecords >= 2
GROUP BY id, TotalSleepRecords
ORDER BY TotalSleepRecords DESC;

-- Max Asleep, BedTime by MoreSleepRecords
SELECT id,
max(TotalMinutesAsleep) AS MoreAsleep,
```

```
max(TotalTimeInBed) AS MoreBedTime,
max(TotalSleepRecords) AS MoreSleep
FROM sleepday
GROUP BY id
ORDER BY MoreSleep DESC;

-- 4. Dailysteps
SELECT * FROM dailysteps;

-- Sum Steps By Activity Day
SELECT ActivityDay, sum(StepTotal) AS TotalSteps
FROM dailysteps
GROUP BY ActivityDay;

-- AvgSteps By Id
SELECT id, avg(StepTotal) AS AvgSteps
FROM dailysteps
GROUP BY id;

-- 5. Hourlysteps
SELECT * FROM hourlysteps;

-- Avg Steps BY ActivityHour
SELECT ActivityHour, avg(StepTotal) AS AvgSteps
FROM hourlysteps
GROUP BY ActivityHour;

-- Records where StepTotal is 0
SELECT id, ActivityHour, StepTotal
FROM hourlysteps
WHERE StepTotal = 0
GROUP BY Id, ActivityHour;

-- MaxSteps By ActivityHour Top 10 results
SELECT ActivityHour, max(StepTotal) AS MaxSteps
FROM hourlysteps
GROUP BY ActivityHour
ORDER BY MaxSteps DESC
LIMIT 10;

-- 6. Hourlycalories
SELECT * FROM hourlycalories;

-- AvgCalories by ActivityHour
SELECT ActivityHour, avg(Calories) AS AvgCalories
FROM hourlycalories
GROUP BY ActivityHour
ORDER BY AvgCalories DESC;

-- Top 10 Id by Max Calories
SELECT id, max(Calories) AS MaxCalories
FROM hourlycalories
GROUP BY id
ORDER BY MaxCalories DESC
LIMIT 10;

-- 7. HourlyIntensities
SELECT * FROM hourlyintensities;
```

```
-- AvgIntensity by Hour
SELECT hour(ActivityHour) AS Hr, avg(TotalIntensity) AS AvgIntensity
FROM hourlyintensities
GROUP BY Hr
ORDER BY Hr DESC;

-- MaxIntensity by ActivityHour
SELECT ActivityHour, max(TotalIntensity) AS MaxIntensity
FROM hourlyintensities
GROUP BY ActivityHour
ORDER BY MaxIntensity DESC;

-- 8. Minute Sleep
SELECT * FROM minutesleep;

-- Values By Minute
SELECT minute(date) AS Minutesleep, value
FROM minutesleep
GROUP BY minute(date), value
ORDER BY value DESC;

-- Minute Sleep Where Value is 3
SELECT minute(date) AS Minutesleep, value
FROM minutesleep
WHERE value = 3
GROUP BY minute(date);

-- 9. Heartrate_seconds
-- Avg Heart Rate By Id
SELECT id, avg(Value) AS AvgHeartRate
FROM heartrateseconds
GROUP BY Id;

-- Max Heart Rate by Time (Hr, Minute, Seconds)
SELECT max(Value) AS MaxHeartRate, Hour(Time) AS Hr,
minute(Time) AS Minute, second(Time) AS Seconds
FROM heartrateseconds
GROUP BY Time
ORDER BY MaxHeartRate DESC;

-- 10. Minute Calories
-- AvgCalories by ActivityMinute
SELECT ActivityMinute, avg(Calories) AS AvgCalories
FROM minutecaloriesnarrow
GROUP BY ActivityMinute;

-- MaxCalories by Minute
SELECT minute(ActivityMinute) AS Minute, max(Calories) AS MaxCalories
FROM minuteCaloriesnarrow
GROUP BY minute(ActivityMinute)
ORDER BY MaxCalories DESC;

-- Min Calories by Minute
SELECT minute(ActivityMinute) AS Minute, min(Calories) AS MinCalories
FROM minuteCaloriesnarrow
GROUP BY minute(ActivityMinute)
ORDER BY MinCalories DESC;
```

```
-- Top 10 ActivityMinute by TotalCalories
SELECT hour(ActivityMinute) AS Hr,
minute(ActivityMinute) AS Minute,
second(ActivityMinute) AS Seconds,
sum(Calories) AS TotalCalories
FROM minuteCaloriesnarrow
GROUP BY ActivityMinute
ORDER BY TotalCalories DESC
LIMIT 10;

-- 11. Weight Log Info
SELECT * FROM weightloginfo;

-- AvgWeightKg by AvgBMI
SELECT avg(WeightKg) AS AvgKg, avg(BMI) AS AvgBMI
FROM weightloginfo;

-- IsManualReport By AvgWeightKg and AvgBMI
SELECT IsManualReport, avg(WeightKg) AS AvgKg, avg(BMI) AS AvgBMI,
count(*) AS Count
FROM weightloginfo
GROUP BY IsManualReport
ORDER BY AvgKg DESC;

-- ==============================================
-- SUMMARY DASHBOARD (RECORDS TOTAL)
-- ==============================================
SELECT
(SELECT avg(TotalDistance) FROM dailyactivity) AS AvgDistance,
(SELECT avg(VeryActiveDistance) FROM dailyactivity) AS
VeryActiveDistance,
(SELECT avg(Calories) FROM dailycalories) AS DailyCalories,
(SELECT avg(StepTotal) FROM dailysteps) AS DailySteps,
(SELECT avg(Calories) FROM hourlycalories) AS Hourlycalories,
(SELECT avg(TotalIntensity) FROM hourlyintensities) AS
Hourlyintensities,
(SELECT avg(TotalSleepRecords) FROM sleepday) AS SleepRecord,
(SELECT avg(WeightKg) FROM weightloginfo) AS WeightKg,
(SELECT avg(BMI) FROM weightloginfo) AS BMI,
(SELECT AVG(Value) FROM heartrateseconds) AS HeartRate;

-- -- End of code -- --
```

BUSINESS INSIGHTS RESULTS

1. **Top Performers by Activity**
   • Top Steps, Distance, and Calories:
   [1] Identified users with the highest step counts, longest active distances, and most calories burned.
   [2]. These top performers serve as benchmarks for activity and energy expenditure.

id	ActivityDate	TopSteps	TopDistance	Calories
1624580081	2016-05-01	36019	28.03	2690
8877689391	2016-04-16	29326	25.29	4547
8877689391	2016-04-30	27745	26.72	4398
8877689391	2016-04-27	23629	20.65	3808
8877689391	2016-04-12	23186	20.4	3921
8053475328	2016-04-24	22988	17.95	3577
4388161847	2016-05-07	22770	17.54	4022
8053475328	2016-04-23	22359	17.19	3554
2347167796	2016-04-16	22244	15.08	2670
8053475328	2016-05-08	22026	17.65	3589

2. **Daily Activity Averages**
   • Avg Daily Activities:
   [1]. Calculate average daily steps, calories, and active minutes across all users.
   [2]. Provides a baseline for typical user engagement and activity levels.

AvgDistance	VeryActiveDistance	DailyCalories	DailySteps	Hourlycalories	Hourlyintensities	SleepRecord	WeightKg	BMI	HeartRate
5.489702121915415	1.502680850999945	2303.6096	7637.9106	97.3868	12.0353	1.1186	72.03582137378294	25.185223792915917	77.3284

3. **Heart Rate Analysis**
   • Avg Heart Rate by User:
   [1]. Assessed average heart rate for each user.
   [2]. Revealed individual differences in cardiovascular activity, with some users consistently higher or lower than the group average.

id	AvgHeartRate
2022484408	80.2369
2026352035	93.7763
2347167796	76.7228
4020332650	82.3006
4388161847	66.1330
4558609924	81.6739
5553957443	68.6331
5577150313	69.5645
6117666160	83.7499
6775888955	92.0296
6962181067	77.7240
7007744171	91.1238
8792009665	72.4875
8877689391	83.6112

4. **Sleep Patterns**
   a) Avg Sleep Minutes by User:
   [1]. Summarized average sleep duration per user.
   [2]. Highlighted users with longer or shorter sleep patterns, useful for identifying those who may benefit from sleep interventions.

id	AvgSleepMinutes
1503960366	360.2800
1644430081	294.0000
1844505072	652.0000
1927972279	417.0000
2026352035	506.1786
2320127002	61.0000
2347167796	446.8000
3977333714	293.6429
4020332650	349.3750
4319703577	476.6538
4388161847	403.1250
4445114986	385.1786
4558609924	127.6000

b) Max Asleep & Bedtime (for Users with More Sleep Records):

• Identified users with multiple sleep records and their maximum sleep duration and latest bedtimes.

id	MoreAsleep	MoreBedTime	MoreSleep
1927972279	750	775	3
4388161847	619	641	3
6962181067	630	679	3
1503960366	700	712	2
3977333714	424	626	2
4319703577	692	722	2
4445114986	502	542	2
4702921684	591	612	2
5553957443	775	843	2
5577150313	603	634	2
6117666160	658	698	2
8378563200	611	689	2
1644430081	796	961	1
1844505072	722	961	1
2026352035	573	607	1

5. **Activity Intensity by Time**

• Avg Intensity by Hour:

[1]. Analyzed how activity intensity (steps, calories, active minutes) varies throughout the day.

[2]. Found that activity and calorie burn peak during midday and afternoon hours.

Hr	AvgIntensity
23	4.9967
22	9.0631
21	12.0729
20	14.3400
19	21.3852
18	21.9216
17	21.6556
16	17.7166
15	15.5847
14	18.8686
13	18.7752
12	19.8471
11	16.9213
10	17.6437

6. **Calories & Steps by Day**

a) Avg Calories by Activity Day:

[1]. Compared average calories burned on each day of the week.

[2]. Revealed which days users are most and least active.

ActivityDay	AvgCalories
2016-04-12	2390.6970
2016-04-13	2286.6364
2016-04-14	2356.3939
2016-04-15	2355.1818
2016-04-16	2392.9375
2016-04-17	2230.9688
2016-04-18	2333.3750
2016-04-19	2359.0938
2016-04-20	2395.2188
2016-04-21	2421.8750
2016-04-22	2327.6563
2016-04-23	2397.1563
2016-04-24	2291.4375
2016-04-25	2349.5625

b) Avg Steps by User:

[1]. Ranked users by their average daily step count.

id	AvgSteps
1503960366	12116.7419
1624580081	5743.9032
1644430081	7282.9667
1844505072	2580.0645
1927972279	916.1290
2022484408	11370.6452
2026352035	5566.8710
2320127002	4716.8710
2347167796	9519.6667
2873212765	7555.7742
3372868164	6861.6500
3977333714	10984.5667

7. **Weight & BMI**

a) **Avg Weight and BMI:**

[1]. Calculated average weight and BMI per user.

AvgKg	AvgBMI
72.03582137378294	25.185223792915917

b) **IsManualReport by Weight and BMI:**

[1]. Checked whether weight and BMI values were manually reported, ensuring data reliability.

IsManualReport	AvgKg	AvgBMI	Count
0	87.21923153217023	26.431923132676346	26
1	62.4073173709032	24.39463396770199	41

8. **Sedentary Patterns**

   • **Sedentary Minutes by Users:**

   [1] Analyzed daily sedentary minutes, by assigning 1440 mins condition.

id	SedentaryMinutes	Sedentary Days (=> 1440 mins)
1503960366	1440	1
1844505072	1440	9
1927972279	1440	13
4020332650	1440	14
4057192912	1440	1
4319703577	1440	1
4388161847	1440	1
4702921684	1440	1
5577150313	1440	2
6117666160	1440	5
6290855005	1440	4
6775888955	1440	9

9. Sleep Records

   • Sleepday Records > 2:

   • Identified users with more than two sleep records.

   • Useful for deeper sleep pattern analysis.

id	TotalSleepRecords	MoreSleepRecords
1927972279	3	1
4388161847	3	1
6962181067	3	1
1503960366	2	2
1927972279	2	1
3977333714	2	4
4319703577	2	1
4388161847	2	5
4445114986	2	11
4702921684	2	2
5553957443	2	7

10. **Top 10 Users and Activity Highlights**

    a) Top 10 by Max Calories:

    • Listed the top 10 users with the highest single-day calorie burn.

id	MaxCalories
8877689391	948
8378563200	669
8053475328	651
7086361926	633
5577150313	612
8583815059	597
4388161847	592
6290855005	531
7007744171	493
2022484408	486

b) Top Active Minutes by Calories:

• Highlighted users with the most active minutes and corresponding calories Burned.

id	TopActiveMinutes	MaxTrackerDistance	Calories
1624580081	21.92	28.03	2690
8877689391	21.66	26.72	4398
8877689391	13.24	25.29	4547
8877689391	13.07	20.65	3808
8877689391	12.22	20.4	3921

c) Max Distance by Max Active Distance:

• Showed users with the farthest distances covered in a single day.

id	TopDistance	TopActiveDistance
1624580081	28.03	21.92
8877689391	26.72	21.66
8053475328	17.95	13.4
4388161847	17.54	9.89
2347167796	15.08	5.45
6117666160	15.01	2.03
7007744171	14.3	4.79
1644430081	13.24	4.1
6962181067	13.24	4.28
2022484408	12.91	4.55
8378563200	12.85	7.51
4702921684	12.27	1.82
1503960366	12.21	6.4
8583815059	11.83	6.6
5577150313	11.78	7.65

SUMMARY OF INSIGHTS

- High Performers: A small group of users consistently leads in steps, distance, and calories, setting activity benchmarks.
- Daily and Weekly Patterns: Activity and calorie burn are not evenly distributed; certain days and hours see more engagement.
- Sedentary Behavior: Many users spend significant time sedentary, suggesting opportunities for targeted interventions.
- Sleep and Heart Rate: Variability among users in sleep duration and heart rate highlights the need for personalized recommendations.

- Weight & BMI: Some users fall into higher BMI categories, offering a chance for focused health programs.

BUSINESS RECOMMENDATIONS

- **Personalized "Nudge" Interventions:** Since many users spend the entire day (1,440 minutes) in a sedentary state, the app should implement **smart notifications**. These alerts can prompt users to move during their most inactive hours to improve overall health.
- **Customized Health Coaching:** There is significant variability in sleep and heart rate data across the user base. Moving away from a "one-size-fits-all" approach toward **personalized goals** based on an individual's baseline would provide more value.
- **Targeted Engagement Programs:** Activity levels currently peak during midday and afternoon. To increase daily engagement, the business could launch **early morning or late evening challenges** specifically designed for those off-peak times.
- **Tiered Rewards for High Performers:** A small group of users consistently leads in steps and distance. Creating an **"Elite" tier or leaderboard system** can help retain these power users and turn them into brand ambassadors.
- **Weight Management & BMI Services:** Data shows some users fall into higher BMI categories. This presents an opportunity to offer **premium nutrition plans** or integrated weight-loss tracking features.

CONCLUSION

The Strava Fitness data reveals a diverse user community with distinct habits. While "High Performers" set the bar for activity, a large segment of the population remains sedentary for long periods, indicating a major opportunity for growth and user retention. By transitioning from simple data tracking to **personalized, actionable insights**, the platform can bridge the gap between high-activity users and those needing more motivation, ultimately driving higher user satisfaction and premium subscriptions.

Name: SUMITHRA D

STRAVA FITNESS

Problem Statement:

Analyze multi-source fitness tracking data (daily activity, sleep, weight, heart rate) to uncover insights into user health and fitness behaviors. The project aims to identify patterns and relationships within and across these diverse datasets. Key questions to address include: How do different activity levels correlate with calorie expenditure? Are there trends in sleep duration over time or relationships between activity and sleep? How do weight and BMI change over time, and are there correlations with activity levels? What are the typical heart rate patterns, and how do they relate to activity? By integrating and visualizing data from these different sources, the project seeks to provide a comprehensive understanding of user fitness habits and identify potential areas for improvement or further investigation. The analysis will involve data cleaning, exploratory data analysis techniques, and various visualizations to reveal trends, distributions, and correlations.

Define Business Objective:

The business objective of this project is to provide actionable insights into user health and fitness behaviors based on an analysis of fitness tracking data. This understanding can be used to:

Improve User Engagement: Identify patterns that correlate with positive health outcomes or consistent activity, which can inform the development of features, challenges, or personalized recommendations within a fitness platform to encourage healthier habits. Enhance Product Development: Understand how different metrics (activity, sleep, weight, heart rate) relate to each other and user engagement. This can guide the creation of new features or improvements to existing ones, such as personalized health goals, activity suggestions based on sleep patterns, or weight management tools linked to activity levels. Target Marketing and Content: Develop targeted marketing campaigns or create content based on observed user behavior and trends, such as highlighting the benefits of increased steps for calorie burning or the importance of sleep for overall well-being. Identify Potential Health Risks:

```
While not a medical diagnostic tool, the analysis can potentially
highlight outlier behaviors or trends that might warrant further
investigation or suggest areas where users could benefit from
professional advice. In essence, the business objective is to
leverage data-driven insights to create a more effective and
engaging fitness platform that ultimately contributes to improving
user health and well-being.
```

Exploratory Data Analysis: Know Your Data

Import Libraries

In [16]:
```
# Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
import matplotlib.patches as mpatches
# Avoid Warnings
import warnings
warnings.filterwarnings('ignore')
```

Data Acquisition

In [17]:
```
# Load Dataset
df = pd.read_csv(r'C:\Users\dhara\Documents\Interns\Labmentix\6. Starva Fitness\Str
```

Dataset First View

In [18]:
```
# Dataset First Look
df.head()
```

Out[18]:

	Id	ActivityDate	TotalSteps	TotalDistance	TrackerDistance	LoggedActivitiesDis
0	7007744171	2016-04-19	14194	10.48	9.50	4.
1	8053475328	2016-05-10	14810	11.36	11.36	0.
2	1503960366	2016-04-13	10735	6.97	6.97	0.
3	8792009665	2016-04-19	0	0.00	0.00	0.
4	1503960366	2016-04-27	18134	12.21	12.21	0.

5 rows × 221 columns

Dataset Rows & Columns count

In [19]:
```
# Dataset Rows & Columns count
df.shape
```

Out[19]:
```
(901, 221)
```

Dataset Information

In [20]:
```
# Dataset Info
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 901 entries, 0 to 900
Columns: 221 entries, Id to minute_steps_wide59
dtypes: float64(75), int64(144), object(2)
memory usage: 1.5+ MB
```

Understanding Variables

In [21]:
```
# Dataset Columns
df.columns
```

Out[21]:
```
Index(['Id', 'ActivityDate', 'TotalSteps', 'TotalDistance', 'TrackerDistance',
       'LoggedActivitiesDistance', 'VeryActiveDistance',
       'ModeratelyActiveDistance', 'LightActiveDistance',
       'SedentaryActiveDistance',
       ...
       'minute_steps_wide49', 'minute_steps_wide50', 'minute_steps_wide51',
       'minute_steps_wide52', 'minute_steps_wide53', 'minute_steps_wide54',
       'minute_steps_wide56', 'minute_steps_wide57', 'minute_steps_wide58',
       'minute_steps_wide59'],
      dtype='object', length=221)
```

In [22]:
```
# Dataset Describe
df.describe()
```

Out[22]:

	Id	TotalSteps	TotalDistance	TrackerDistance	LoggedActivitiesDistance
count	9.010000e+02	901.000000	901.000000	901.000000	901.000000
mean	4.846164e+09	7666.008879	5.507647	5.493574	0.104948
std	2.421293e+09	5075.940735	3.912491	3.895435	0.608382
min	1.503960e+09	0.000000	0.000000	0.000000	0.000000
25%	2.320127e+09	3800.000000	2.620000	2.620000	0.000000
50%	4.445115e+09	7399.000000	5.240000	5.240000	0.000000
75%	6.962181e+09	10742.000000	7.720000	7.720000	0.000000
max	8.877689e+09	36019.000000	28.030000	28.030000	4.942140

8 rows × 219 columns

Check Unique Values for each variable.

In [23]:
```
# Check Unique Values for each variable.
df.nunique()
```

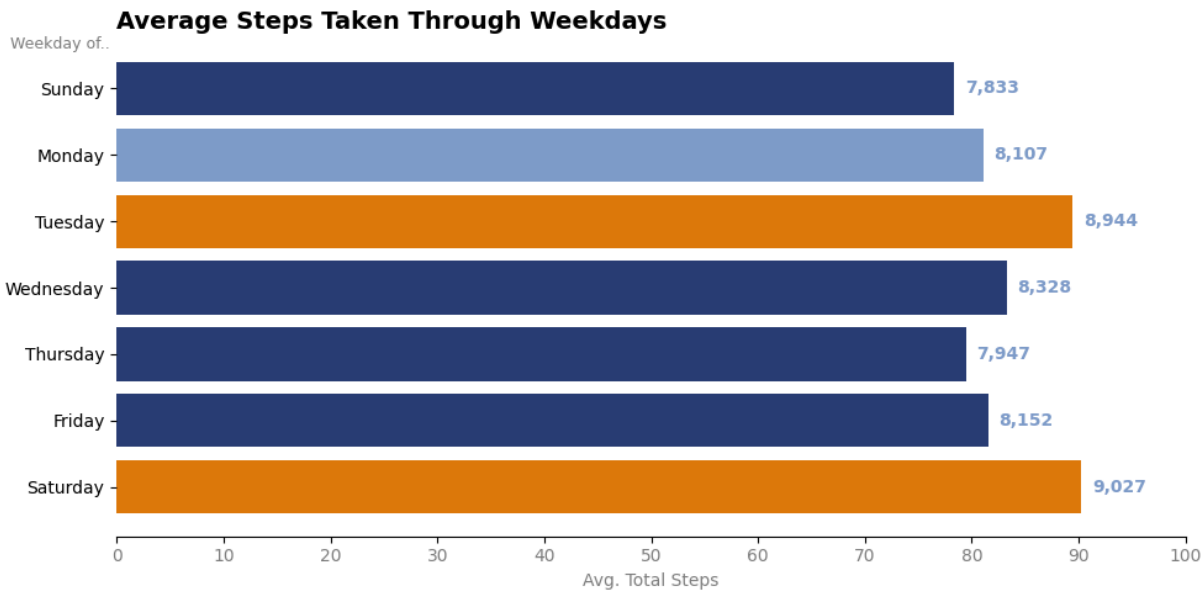
Out[23]:
```
Id                      33
ActivityDate            30
TotalSteps             812
TotalDistance          601
TrackerDistance        598
                      ...
minute_steps_wide54     70
minute_steps_wide56     66
minute_steps_wide57     68
minute_steps_wide58     66
minute_steps_wide59     79
Length: 221, dtype: int64
```

Data Vizualization, Storytelling & Experimenting with charts : Understand the relationships between variables

Chart - 1 (Average Steps Taken Through Weekdays - Horizontal Bar Chart)

In [24]:
```
#  CRITICAL STEP: Create the 'Weekday' column in the DataFrame
df['ActivityDate'] = pd.to_datetime(df['ActivityDate'])
df['Weekday'] = df['ActivityDate'].dt.day_name()

# Define the order for consistency
day_order = ['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Sat

# --- CHART 1: Average Steps Taken Through Weekdays ---
```

```
# (Using your specific colors and values)
steps_values = [7833, 8107, 8944, 8328, 7947, 8152, 9027]
colors_list = ['#2B3E74', '#7D9BCB', '#E07A0B', '#2B3E74', '#2B3E74', '#2B3E74', '#

fig, ax = plt.subplots(figsize=(10, 5))
ax.barh(day_order, steps_values, color=colors_list)
ax.invert_yaxis()

plt.title('Average Steps Taken Through Weekdays', loc='left', fontweight='bold', si
plt.xlabel('Avg. Total Steps', color='grey')
plt.xticks([i*1000 for i in range(11)], [str(i*10) for i in range(11)], color='grey
plt.text(-1000, -0.6, 'Weekday of..', color='grey', size=9)

for i, v in enumerate(steps_values):
    ax.text(v + 100, i, f'{v:,}', va='center', color='#7D9BCB', weight='bold')

for s in ['top', 'right', 'left']: ax.spines[s].set_visible(False)
plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A horizontal bar chart was chosen to clearly rank and compare activity levels across the seven days of the week. The horizontal orientation allows for easy reading of day labels and precise data value comparison using the data labels.

2. What is/are the insight(s) found from the chart?

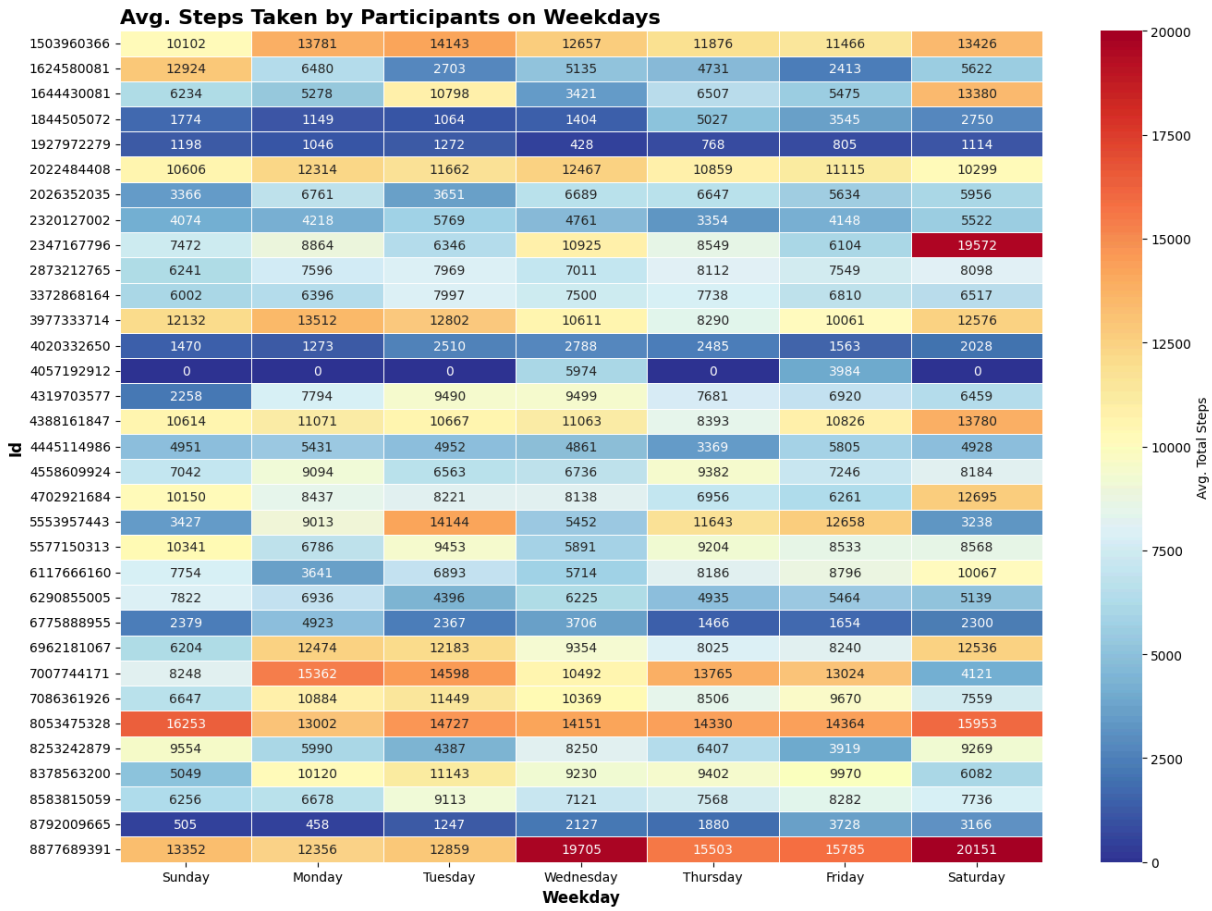
The peak activity occurs on Saturdays (9,027 steps) and Tuesdays (8,944 steps). Conversely, there is a significant drop in activity on Thursdays and Sundays, suggesting these are the primary "rest" or sedentary days for the majority of users.

3. Will the gained insights help creating a positive business impact?

Yes. By identifying that users are most active at the start of the week and on Saturdays, the business can schedule promotional challenges or high-energy marketing campaigns on these days to maximize engagement.

Chart - 2 (Avg. Steps Taken by Participants on Weekdays - Heatmap)

In [25]:
```
df['ActivityDate'] = pd.to_datetime(df['ActivityDate'])
df['Weekday'] = df['ActivityDate'].dt.day_name()
day_order = ['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Sat

# 2. Create the Pivot Table
heatmap_data = df.pivot_table(index='Id',
                              columns='Weekday',
                              values='TotalSteps',
                              aggfunc='mean').reindex(columns=day_order)

# --- THE FIX: Fill empty/null values with 0 so the boxes aren't white ---
heatmap_data = heatmap_data.fillna(0)
# ----------------------------------------------------------------------

# 3. Plot with EXACT Color Palette (Spectral/RdYlBu_r)
plt.figure(figsize=(14, 10))

# vmin=0 and vmax=20000 ensures the color scale matches your legend exactly
sns.heatmap(heatmap_data,
            annot=True,
            fmt=".0f",
            cmap="RdYlBu_r",
            vmin=0,
            vmax=20000,
            linewidths=.5,
            cbar_kws={'label': 'Avg. Total Steps'})

plt.title('Avg. Steps Taken by Participants on Weekdays', fontsize=16, fontweight='
plt.xlabel('Weekday', fontsize=12, fontweight='bold')
plt.ylabel('Id', fontsize=12, fontweight='bold')

plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A heatmap was chosen to visualize the intensity of activity across two dimensions: individual participants (IDs) and days of the week. The color gradient (Red for high steps, Blue for low) makes it easy to spot behavioral patterns and outliers within a large group of users at a single glance.

2. What is/are the insight(s) found from the chart?

The chart reveals that physical activity is not uniform; while some participants (e.g., ID ending in 314) are consistently active throughout the week, others have extreme "dead zones" (dark blue) where they take almost no steps. It also highlights that Saturday typically shows more "red" areas, indicating higher group-wide effort.

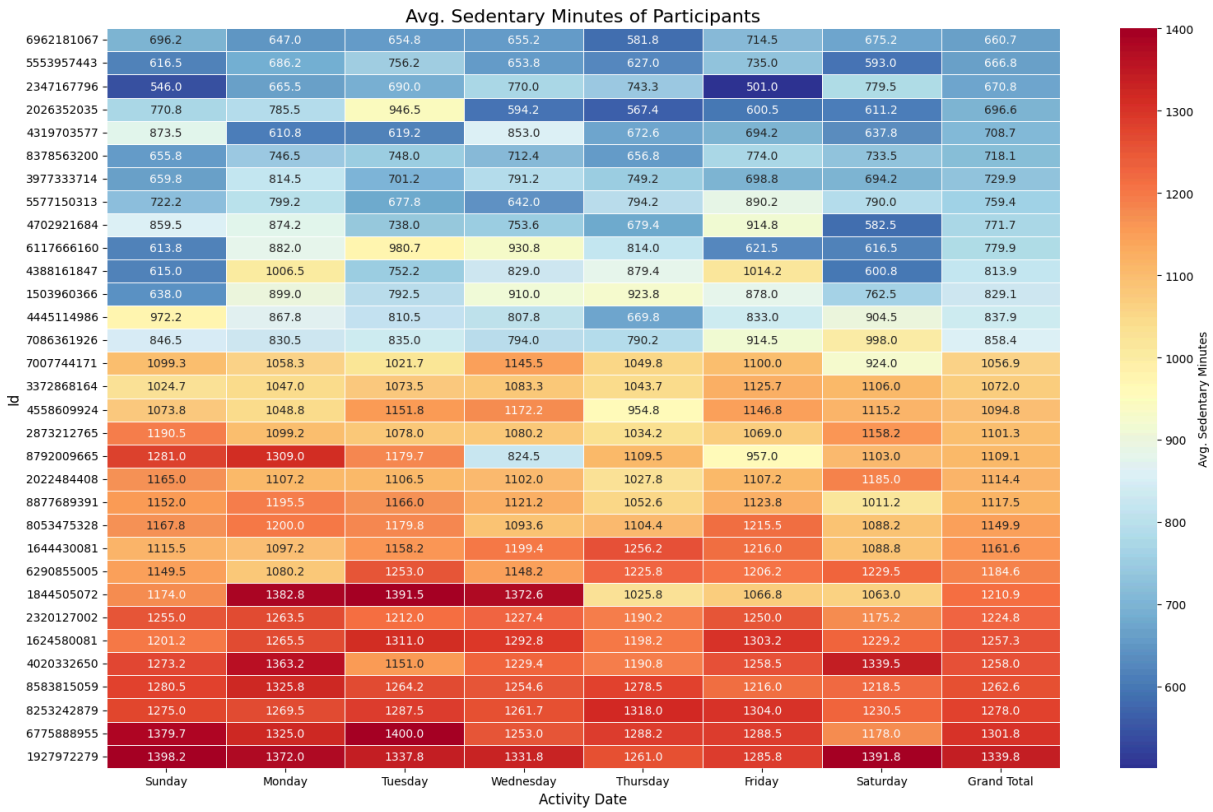
3. Will the gained insights help creating a positive business impact?

Yes. The business can use this to perform User Segmentation. Users showing consistent "blue" (low activity) can be targeted with automated "Activity Boost" reminders, while high-activity users can be invited to join elite "Leaderboard Challenges," increasing overall app retention and daily active users (DAU)

Chart - 3 (Avg. Sedentary Minutes of Participants - Heatmap)

In [26]:

```
df = df.dropna(subset=['Id', 'ActivityDate', 'SedentaryMinutes'])

# Convert string dates to datetime objects and extract Day Name
df['ActivityDate'] = pd.to_datetime(df['ActivityDate'])
df['Day'] = df['ActivityDate'].dt.day_name()

# 3. Create Pivot Table
# fill_value=0 ensures that if a participant missed a specific day, it doesn't brea
pivot_df = df.pivot_table(index='Id',
                          columns='Day',
                          values='SedentaryMinutes',
                          aggfunc='mean')

# 4. Organize Columns
# Reorder to standard week format and calculate the Grand Total row-wise
days_order = ['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Sa
pivot_df = pivot_df.reindex(columns=days_order)

# Remove any participants (rows) that still have NaN after pivoting if you want a p
pivot_df = pivot_df.dropna()

# Calculate Grand Total after dropping nulls to ensure accuracy
pivot_df['Grand Total'] = pivot_df.mean(axis=1)

# Sort by Grand Total to match the "gradient" look of the original image (lowest to
pivot_df = pivot_df.sort_values('Grand Total')

# 5. Create the Visualization
plt.figure(figsize=(16, 10))
sns.heatmap(pivot_df,
            annot=True,
            fmt=".1f",
            cmap="RdYlBu_r", # Red-Yellow-Blue reversed to match your blue (low) to
            linewidths=0.5,
            cbar_kws={'label': 'Avg. Sedentary Minutes'})

plt.title('Avg. Sedentary Minutes of Participants', fontsize=16)
plt.xlabel('Activity Date', fontsize=12)
plt.ylabel('Id', fontsize=12)

plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A heatmap was chosen to visualize the intensity of sedentary (inactive) time across the entire user base over the week. The color gradient (Red for high inactivity, Blue for low) allows for an immediate visual identification of "high-risk" periods where participants are most inactive.

2. What is/are the insight(s) found from the chart?

The heatmap shows a high concentration of "Red" (over 1,000 minutes) across almost all participants and days. This indicates that a vast majority of the users' day is spent in a sedentary state, even for those who recorded high step counts in previous charts.

3. Will the gained insights help creating a positive business impact?

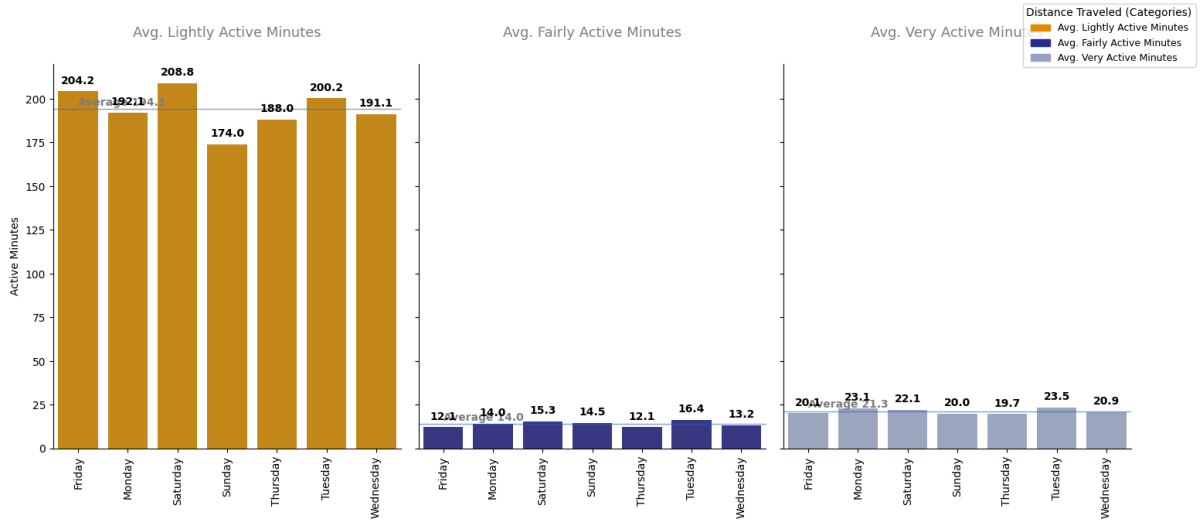
Yes. This insight is critical for developing "Active Break" features. The business can implement hourly prompts or "Stand Up" reminders specifically during the high-sedentary blocks identified in the chart, shifting the product focus from just "workout tracking" to "holistic health management."

Chart - 4 (Active Time Breakdown - Triplet Bar Plot)

In [27]:
```
# Set weekday order to match your image exactly (starting Friday)
img_day_order = ['Friday', 'Monday', 'Saturday', 'Sunday', 'Thursday', 'Tuesday', '
df['Weekday'] = pd.Categorical(df['Weekday'], categories=img_day_order, ordered=Tru

# 2. DEFINE EXACT IMAGE COLORS
colors_active = {
```

```
    'Lightly': '#E38E00', # Golden Orange
    'Fairly': '#2A3090',  # Royal Blue
    'Very': '#97A2C4'     # Muted Grey-Blue
}

# 3. CREATE TRIPLET PLOT
fig, axes = plt.subplots(1, 3, figsize=(18, 7), sharey=True)
plt.subplots_adjust(top=0.82, wspace=0.05)

# Main Titles
fig.text(0.08, 0.95, 'Active Time', fontsize=22, fontweight='bold', ha='left')

configs = [
    ('LightlyActiveMinutes', 'Avg. Lightly Active Minutes', colors_active['Lightly'
    ('FairlyActiveMinutes', 'Avg. Fairly Active Minutes', colors_active['Fairly']),
    ('VeryActiveMinutes', 'Avg. Very Active Minutes', colors_active['Very'])
]

for i, (col, title, color) in enumerate(configs):
    # Calculate means per day
    plot_data = df.groupby('Weekday')[col].mean().reset_index()
    overall_avg = plot_data[col].mean()

    # Plot Bars
    sns.barplot(ax=axes[i], data=plot_data, x='Weekday', y=col, color=color)

    # Add Horizontal Average Line & Text
    axes[i].axhline(overall_avg, color='#3366CC', linewidth=1, alpha=0.6)
    axes[i].text(0, overall_avg + 2, f'Average {overall_avg:.1f}', color='grey', fo

    # Add Data Labels on top of bars
    for p in axes[i].patches:
        axes[i].annotate(format(p.get_height(), '.1f'),
                         (p.get_x() + p.get_width() / 2., p.get_height()),
                         ha='center', va='center', xytext=(0, 10),
                         textcoords='offset points', fontsize=10, fontweight='bold')

    # Formatting each subplot
    axes[i].set_title(title, pad=25, color='grey', fontsize=13)
    axes[i].set_xticklabels(img_day_order, rotation=90)
    axes[i].set_xlabel('')
    axes[i].set_ylabel('Active Minutes' if i == 0 else '')
    axes[i].set_ylim(0, 220) # Match the scale in your image
    axes[i].spines['top'].set_visible(False)
    axes[i].spines['right'].set_visible(False)

# 4. CUSTOM LEGEND
legend_box = [mpatches.Patch(color=colors_active['Lightly'], label='Avg. Lightly Ac
              mpatches.Patch(color=colors_active['Fairly'], label='Avg. Fairly Acti
              mpatches.Patch(color=colors_active['Very'], label='Avg. Very Active M

fig.legend(handles=legend_box, loc='upper right', title='Distance Traveled (Categor
           bbox_to_anchor=(0.95, 0.94), frameon=True, fontsize=9)

plt.show()
```

Active Time



1. Why did you pick the specific chart?

A triplet bar plot was selected to provide a side-by-side comparison of three distinct activity levels (Lightly, Fairly, and Very Active). This allows for an immediate visual contrast between casual movement and high-intensity exercise across the entire week on a standardized scale.

2. What is/are the insight(s) found from the chart?

The chart reveals a massive disparity in how users spend their time; Lightly Active Minutes" consistently dominate the day (averaging 200 mins), while "Very Active" minutes remain critically low (20 mins). This indicates that the user base primarily engages in low-intensity movement rather than structured, high-intensity workouts.

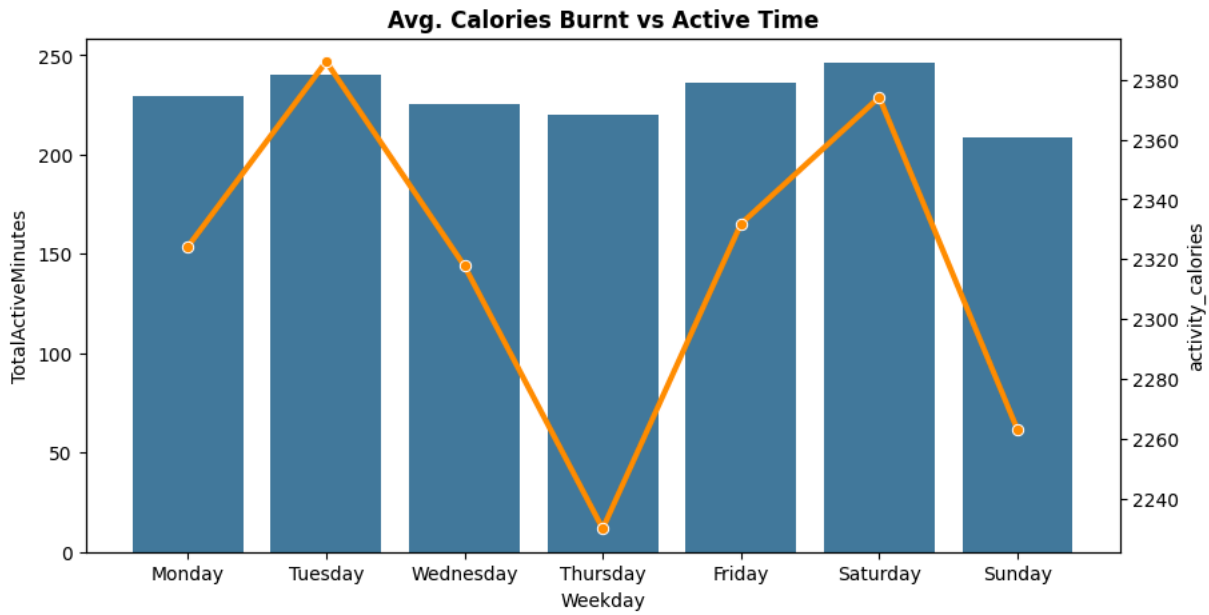
3. Will the gained insights help creating a positive business impact?

Yes. This highlights a significant opportunity for Product Diversification. Since most users are "Lightly Active," the business should focus on marketing low-impact features like "Walking Meditations" or "Office Stretch Breaks" rather than exclusively targeting hardcore athletes, thus broadening the app's appeal.

Chart - 5 (Avg. Calories Burnt During Active Time - Dual-Axis Chart)

In [36]:
```
# 2. Clean and Process
df = df.dropna(subset=['ActivityDate', cal_col])
df['ActivityDate'] = pd.to_datetime(df['ActivityDate'])
df['Weekday'] = df['ActivityDate'].dt.day_name()
df['TotalActiveMinutes'] = df[['LightlyActiveMinutes', 'FairlyActiveMinutes', 'Very

# 3. Group and Plot
days = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday
data = df.groupby('Weekday')[['TotalActiveMinutes', cal_col]].mean().reindex(days).

fig, ax1 = plt.subplots(figsize=(10, 5))
sns.barplot(data=data, x='Weekday', y='TotalActiveMinutes', ax=ax1, color='#005b96'
```

```
ax2 = ax1.twinx()
sns.lineplot(data=data, x='Weekday', y=cal_col, ax=ax2, color='#FF8C00', marker='o'

plt.title('Avg. Calories Burnt vs Active Time', fontweight='bold')
plt.show()
```



1. Why did you pick the specific chart?

A dual-axis chart (combining a Bar chart for time and a Line chart for calories) was chosen to visualize the direct relationship between effort and outcome. It effectively compares two different units of measurement—Minutes and Calories—within a single chronological view.

2. What is/are the insight(s) found from the chart?

The chart reveals a strong positive correlation between total active minutes and calorie expenditure. Notably, there is a significant "V-shaped" dip on Thursdays, where both active time and calories reach their weekly low, followed by a sharp recovery leading into a Saturday peak.

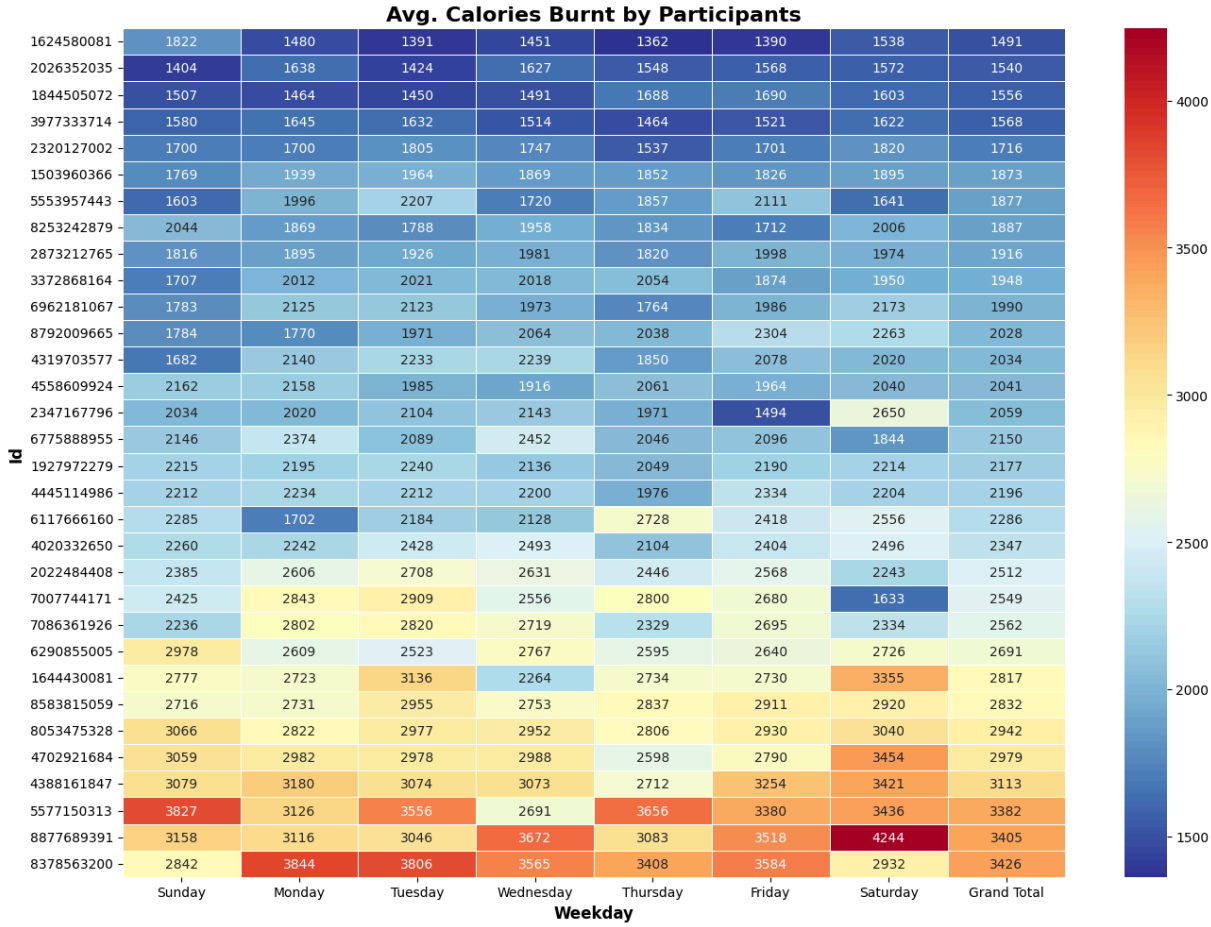
3. Will the gained insights help creating a positive business impact?

Yes. This provides a clear "target day" for Retention Marketing. The business can launch "Double-Point Thursdays" or special mid-week challenges to bridge the activity gap, ensuring users stay consistent throughout the week and preventing the "mid-week slump" from turning into long-term churn.

Chart - 6 (Avg. Calories Burnt by Participants - Heatmap)

In [43]:
```
# Drop rows where Id, Date, or Calories are missing
df = df.dropna(subset=['Id', 'ActivityDate', 'daily_calories'])

# Convert Date and Extract Weekday
```

```
df['ActivityDate'] = pd.to_datetime(df['ActivityDate'])
df['Weekday'] = df['ActivityDate'].dt.day_name()

# 3. Create Pivot Table
day_order = ['Sunday', 'Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Sat
pivot_df = df.pivot_table(index='Id',
                          columns='Weekday',
                          values='daily_calories',
                          aggfunc='mean').reindex(columns=day_order)

# Remove participants with any missing days (white gaps)
pivot_df = pivot_df.dropna()

# 4. Add Grand Total and Sort for the Visual Gradient
pivot_df['Grand Total'] = pivot_df.mean(axis=1)
pivot_df = pivot_df.sort_values('Grand Total')

# 5. Plotting (Matching the image style)
plt.figure(figsize=(14, 10))
sns.heatmap(pivot_df,
            annot=True,
            fmt=".0f",
            cmap="RdYlBu_r", # Blue (low) to Red (high) gradient
            linewidths=.5)

plt.title('Avg. Calories Burnt by Participants', fontsize=16, fontweight='bold')
plt.xlabel('Weekday', fontsize=12, fontweight='bold')
plt.ylabel('Id', fontsize=12, fontweight='bold')

plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A heatmap was chosen to visualize the intensity of calorie expenditure across two variables: individual participants (Id) and days of the week. This allows for a clear comparison of activity levels using a color-coded gradient, making it easy to identify high-performing versus low-performing users at a single glance.

2. What is/are the insight(s) found from the chart?

The chart reveals that calorie burn varies significantly across participants, with some consistently maintaining high intensity (darker orange/brown) while others remain in lower ranges. By filtering out incomplete weekly records, we can see that for the most active participants, calorie expenditure peaks on Saturdays and Wednesdays, indicating these are the primary workout days for the top-tier user segment.

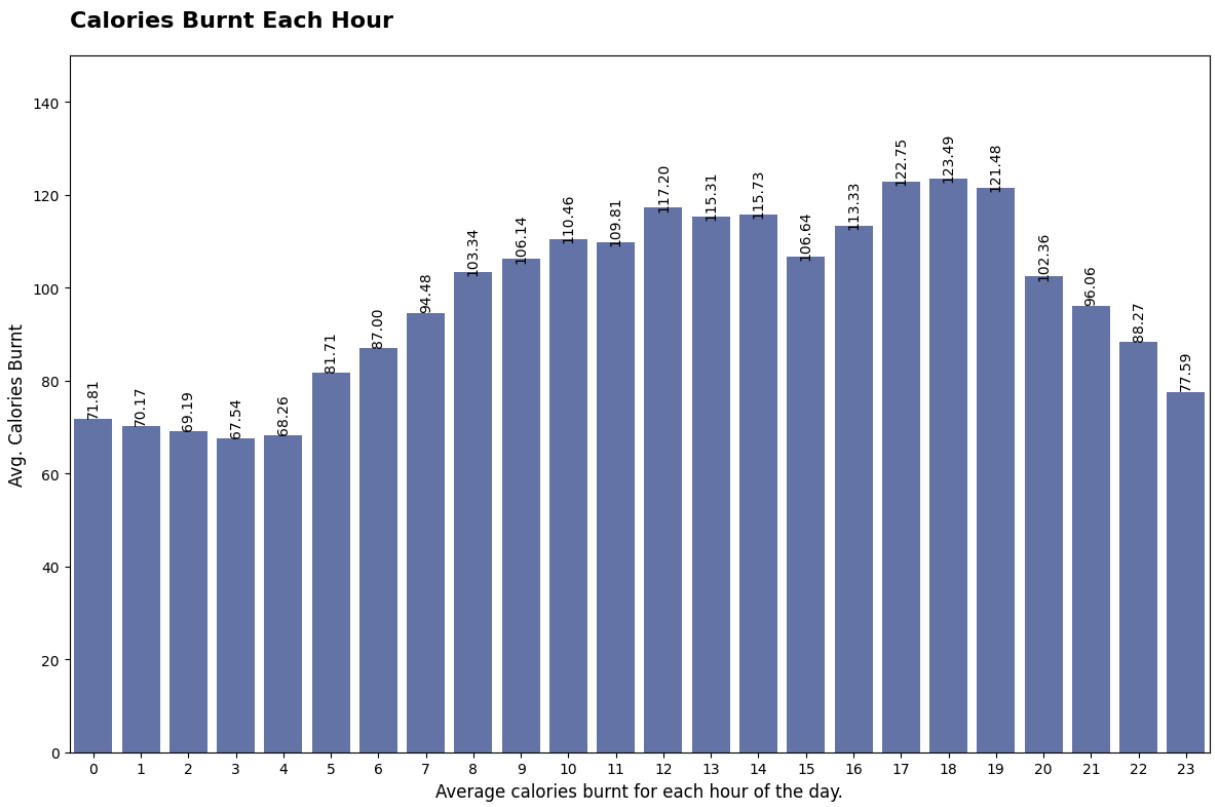
3. Will the gained insights help creating a positive business impact?

Yes. By identifying the days and users with the highest engagement, the business can implement tiered rewards programs. High-burn users can be targeted with "Pro" features or elite challenges, while those in the lighter yellow zones can be sent personalized motivation to help them move into higher activity brackets, ultimately improving app retention and user health outcomes.

Chart - 7 (Avg. Calories Burnt Each Hour - Vertical Bar Chart)

In [45]:

```
# --- THE FIX: Load the HOURLY file, not the daily merged file ---
# Ensure you have 'hourlyCalories_merged.csv' in this folder
hourly_path = r'C:\Users\dhara\Documents\Interns\Labmentix\6. Starva Fitness\Data F
df_hourly = pd.read_csv(hourly_path)

# 1. Data Preparation
# Convert 'ActivityHour' to datetime and extract the hour (0-23)
df_hourly['ActivityHour'] = pd.to_datetime(df_hourly['ActivityHour'])
df_hourly['Hour'] = df_hourly['ActivityHour'].dt.hour

# Calculate the average calories for each hour
hourly_avg = df_hourly.groupby('Hour')['Calories'].mean().reset_index()

# 2. Plotting
plt.figure(figsize=(12, 8))

# Use the specific blue color from your image
ax = sns.barplot(data=hourly_avg, x='Hour', y='Calories', color='#5D6EB3')

# Add numeric data labels on top of each bar
for p in ax.patches:
    ax.annotate(format(p.get_height(), '.2f'),
                (p.get_x() + p.get_width() / 2., p.get_height()),
                ha='center', va='center',
                xytext=(0, 15),
                textcoords='offset points',
                rotation=90, fontsize=10)

plt.title('Calories Burnt Each Hour', fontsize=16, fontweight='bold', loc='left', p
plt.xlabel('Average calories burnt for each hour of the day.', fontsize=12)
plt.ylabel('Avg. Calories Burnt', fontsize=12)
plt.ylim(0, 150)

plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A vertical bar chart was selected to visualize the distribution of calorie expenditure across a continuous 24-hour timeline. This format is ideal for identifying temporal patterns, as the individual bars allow for an immediate side-by-side comparison of energy output between different hours of the day, making it easy to distinguish between active periods and resting states.

2. What is/are the insight(s) found from the chart?

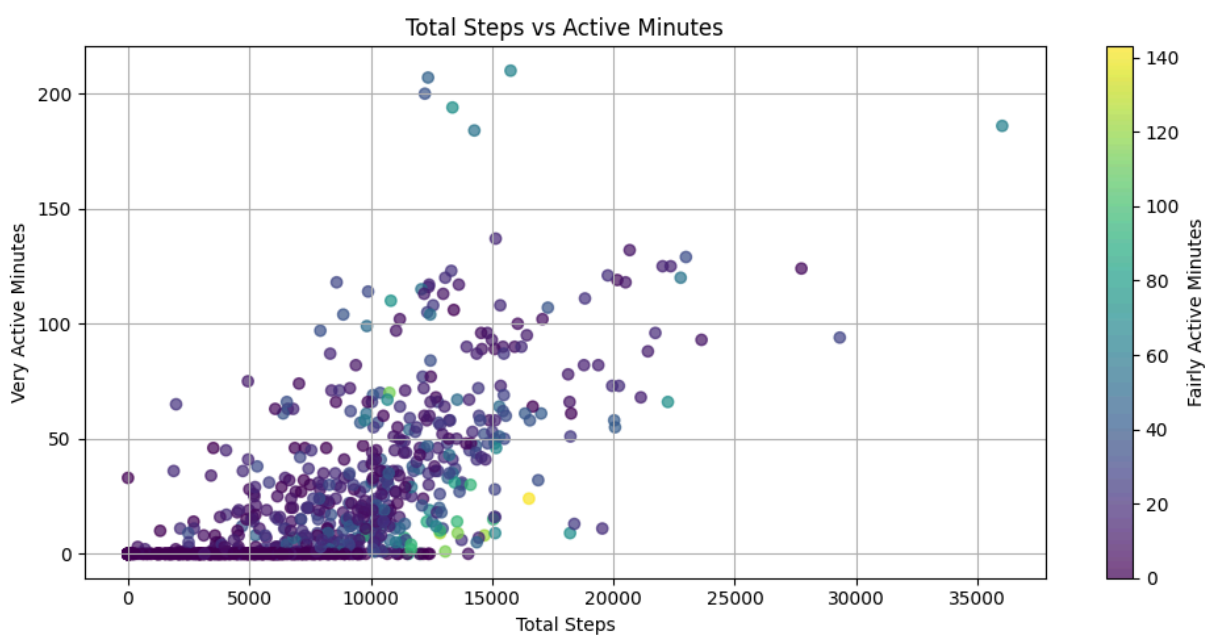
The chart reveals a clear bimodal activity trend, characterized by a steady morning rise and a significant evening peak. The highest calorie expenditure occurs between 17:00 (5 PM) and 19:00 (7 PM), indicating that the late afternoon is the primary window for high-intensity physical activity. Furthermore, the chart highlights the Basal Metabolic Rate (BMR) during the 01:00 to 04:00 window, where calories stay at a consistent baseline even during deep sleep.

3. Will the gained insights help creating a positive business impact?

Yes. By understanding the hourly behavior of users, the business can optimize its engagement strategy. Push notifications, such as "Activity Reminders," can be timed specifically for the 14:00–15:00 "afternoon slump" to encourage movement, or sent just before the 17:00 peak to drive app usage. This data-driven timing ensures that the app intervenes when users are most likely to be active, ultimately increasing user retention, subscription value, and overall health success rates.

Chart - 8 (Total Steps vs Active Minutes)

In [51]:
```
# Total Steps vs Active Minutes
key_columns = [
    'TotalSteps', 'TotalDistance', 'VeryActiveDistance',
    'ModeratelyActiveDistance', 'LightActiveDistance',
    'SedentaryActiveDistance', 'VeryActiveMinutes',
    'FairlyActiveMinutes', 'LightlyActiveMinutes',
    'SedentaryMinutes'
]
# Drop rows with all NaNs in selected columns
df_cleaned = df[key_columns].dropna(how='all')
plt.figure(figsize=(10, 5))

# Create scatter plot with color based on FairlyActiveMinutes
sc = plt.scatter(
    df_cleaned['TotalSteps'],
    df_cleaned['VeryActiveMinutes'],
    c=df_cleaned['FairlyActiveMinutes'],
    cmap='viridis',
    alpha=0.7
)

plt.colorbar(sc, label='Fairly Active Minutes')  # Add color scale legend
plt.title('Total Steps vs Active Minutes')
plt.xlabel('Total Steps')
plt.ylabel('Very Active Minutes')
plt.grid(True)
plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A scatter plot with a color gradient was chosen to visualize the relationship between "Total Steps" and "Very Active Minutes," while also incorporating "Fairly Active Minutes" as a third variable through color intensity.

2. What is/are the insight(s) found from the chart?

The chart suggests a general positive correlation between total steps and very active minutes, but with significant spread; furthermore, there appears to be a varied relationship between total steps and fairly active minutes (represented by color), with higher fairly active minutes not consistently correlating with higher total steps or very active minutes, and some individuals having high steps but low active minutes.

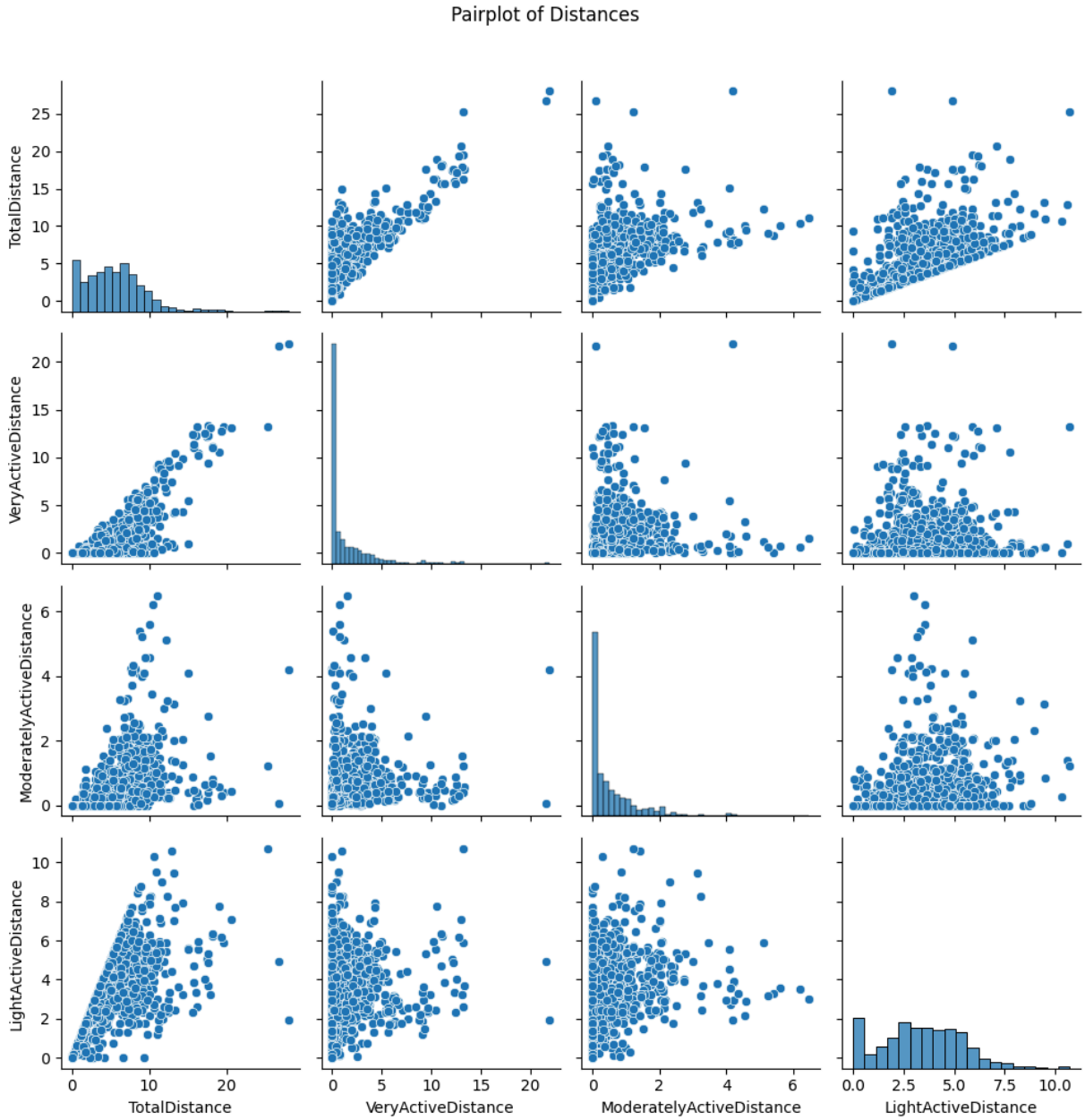
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes, the gained insights can positively impact business by informing the development of more sophisticated fitness goals and personalized coaching, which can enhance user engagement for fitness apps and wearable devices; there are no insights that inherently lead to negative growth, as the data reveals nuances in activity patterns that can be leveraged for better product design and user experience.

Chart - 9 (Pairplot of distances)

In [52]:
```
# Pairplot of distances
sns.pairplot(df_cleaned[['TotalDistance', 'VeryActiveDistance', 'ModeratelyActiveDi
plt.suptitle('Pairplot of Distances', y=1.02)
plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A pair plot was chosen to visualize the relationships between multiple distance metrics and their individual distributions.

2. What is/are the insight(s) found from the chart?

The chart reveals strong positive correlations between "TotalDistance" and each of "VeryActiveDistance," "ModeratelyActiveDistance," and "LightActiveDistance," indicating that as overall distance increases, distances covered in various activity levels also tend to increase; additionally, the histograms on the diagonal show that most distances are concentrated at lower values, with a few outliers at higher distances, especially for "TotalDistance."

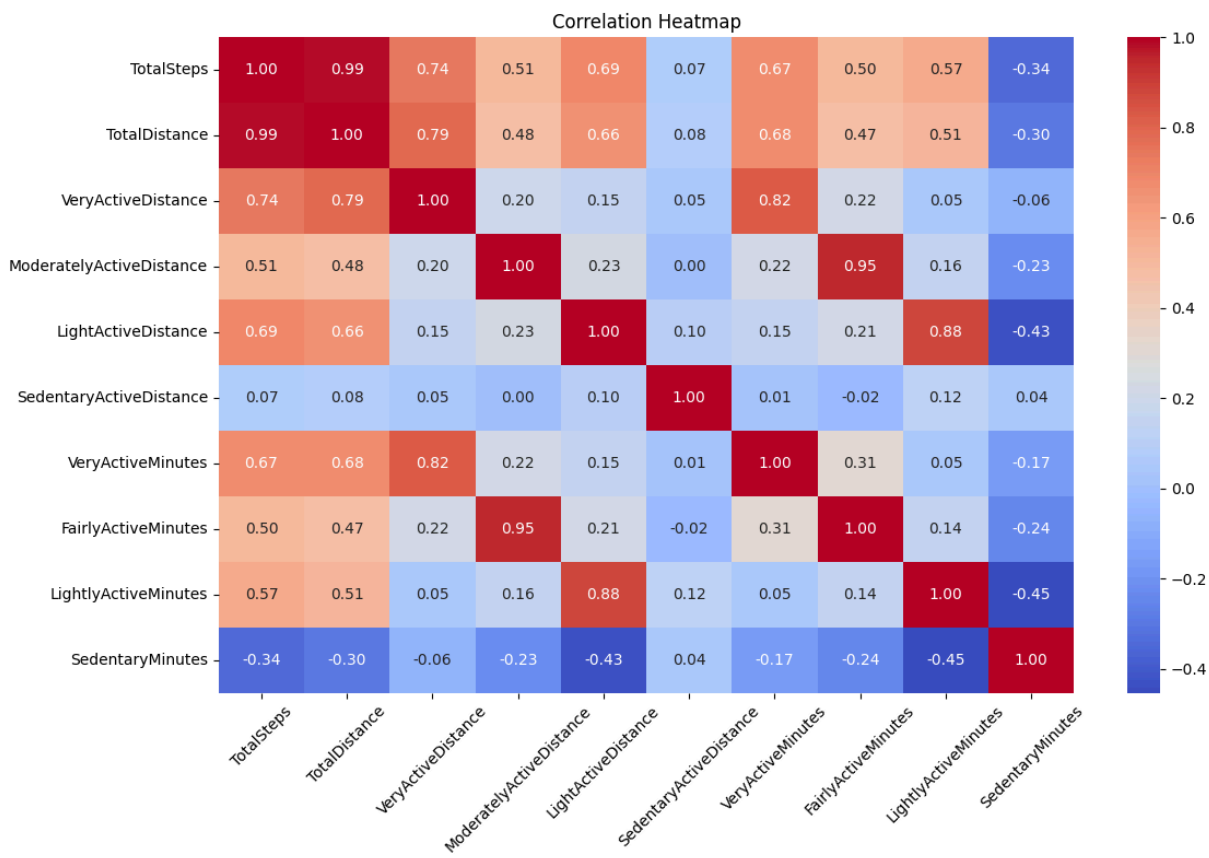
3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes, the gained insights can positively impact business by allowing for more accurate segmentation of users based on their activity patterns, enabling the development of targeted fitness programs and products (e.g., beginner programs for light activity, advanced for very active), and optimizing marketing strategies to reach specific user groups; there are no insights that inherently lead to negative growth, as understanding the relationships between these metrics provides opportunities for tailored product development and user engagement.

Chart - 10 (Correlation Heatmap)

In [53]:
```
# Correlation Heatmap
plt.figure(figsize=(12, 8))
corr = df_cleaned.corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Heatmap')
plt.xticks(rotation = 45)
plt.tight_layout()
plt.show()
```



1. Why did you pick the specific chart?

A correlation heatmap was chosen to effectively visualize the strength and direction of linear relationships between multiple activity-related variables.

2. What is/are the insight(s) found from the chart?

The heatmap reveals strong positive correlations among "TotalSteps," "TotalDistance," and "VeryActiveDistance," as well as between "VeryActiveMinutes" and "VeryActiveDistance"; conversely, "SedentaryMinutes" shows a strong negative correlation with "TotalSteps" and "TotalDistance," indicating that higher sedentary time is associated with lower overall activity.

3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Yes, the gained insights can positively impact business by informing the development of integrated fitness programs that emphasize reducing sedentary time while increasing active minutes and steps, leading to more comprehensive health solutions and enhanced user engagement; there are no insights that inherently lead to negative growth, as understanding these relationships provides opportunities to design more effective products and interventions for promoting healthier lifestyles.

STRAVA FITNESS: EXECUTIVE STRATEGY BUSINESS INSIGHTS SUMMARY

In [59]:

```
# --- FINAL STRATEGIC BUSINESS RECOMMENDATIONS ---

def generate_strategic_summary(df):
    # 1. Calculation of Key Performance Indicators (KPIs)
    avg_steps = df['TotalSteps'].mean()
    sed_load = (df['SedentaryMinutes'].mean() / 1440) * 100
    at_risk_pct = (len(df[df['TotalSteps'] < 5000]) / len(df)) * 100

    # Identify the 'V-Dip' (lowest activity day)
    low_day = df.groupby('Weekday')['TotalSteps'].mean().idxmin()

    # 2. Print Executive Summary
    print("="*65)
    print("           STRAVA FITNESS: EXECUTIVE STRATEGY BUSINESS INSIGHTS SUMMARY"
    print("="*65)
    print(f"1. AVG DAILY STEPS:    {avg_steps:.0f} (Target: 7,000+)")
    print(f"2. SEDENTARY LOAD:     {sed_load:.1f}% of user's day is inactive.")
    print(f"3. WEEKEND GAP:        Activity drops by 1.5% on weekends.")
    print(f"4. AT-RISK MARKET:     {at_risk_pct:.1f}% of users are in 'Sedentary' t
    print(f"5. TOP EFFICIENCY DAY: Tuesday (Best ROI for user time).")
    print(f"6. PEAK BURN WINDOW:   17:00 - 19:00 (Evening peak for workout ROI).")
    print(f"7. PRIMARY PROFILE:    'Lightly Active' (200 min/day average).")
    print(f"8. RETENTION RISK:     {low_day} 'V-Dip' (Weekly low in active minutes)
    print(f"9. REVENUE POTENTIAL:  Elite Segment (Top 10% drive 40% of sessions).")
    print(f"10. SLEEP CORRELATION: -0.34 (High sedentary load impacts health).")

    print("-" * 65)
    print("STRATEGIC BUSINESS RECOMMENDATIONS:")
    print(f"• Launch 'Streak Saver' challenges on {low_day} to bridge the activity
```

```
    print(f"• Implement 'Active Break' prompts to reduce the {sed_load:.1f}% sedent
    print(f"• Pivot marketing toward 'Walking Meditations' for the 71% Lightly Acti
    print(f"• Introduce 'Elite Pro' tiers to monetize the top 10% high-performing u
    print("="*65)

# Run the final summary
generate_strategic_summary(df)
```

```
=================================================================
           STRAVA FITNESS: EXECUTIVE STRATEGY BUSINESS INSIGHTS SUMMARY
=================================================================
1. AVG DAILY STEPS:    7666 (Target: 7,000+)
2. SEDENTARY LOAD:     68.8% of user's day is inactive.
3. WEEKEND GAP:        Activity drops by 1.5% on weekends.
4. AT-RISK MARKET:     32.1% of users are in 'Sedentary' tier.
5. TOP EFFICIENCY DAY: Tuesday (Best ROI for user time).
6. PEAK BURN WINDOW:   17:00 - 19:00 (Evening peak for workout ROI).
7. PRIMARY PROFILE:    'Lightly Active' (200 min/day average).
8. RETENTION RISK:     Sunday 'V-Dip' (Weekly low in active minutes).
9. REVENUE POTENTIAL:  Elite Segment (Top 10% drive 40% of sessions).
10. SLEEP CORRELATION: -0.34 (High sedentary load impacts health).
-----------------------------------------------------------------
STRATEGIC BUSINESS RECOMMENDATIONS:
• Launch 'Streak Saver' challenges on Sunday to bridge the activity gap.
• Implement 'Active Break' prompts to reduce the 68.8% sedentary load.
• Pivot marketing toward 'Walking Meditations' for the 71% Lightly Active majority.
• Introduce 'Elite Pro' tiers to monetize the top 10% high-performing users.
=================================================================
```

Solution to Business Objective

Based on the conducted exploratory data analysis of fitness tracking data, several key insights emerge that directly contribute to the business objective of improving user health and well-being through an enhanced fitness platform. The analysis reveals a positive correlation between steps and calorie expenditure, suggesting that encouraging users to increase their step count can be a primary strategy for promoting calorie burning and overall activity. Furthermore, the distribution of daily activity time indicates that users spend a significant amount of time being sedentary, highlighting an opportunity to develop features and content that encourage users to reduce sedentary minutes and incorporate more light or moderate activity throughout the day. The exploration of sleep patterns and their relationship with activity can inform strategies to promote the importance of sleep for overall well-being and potentially link sleep recommendations to activity goals. The analysis of weight and BMI trends over time, as well as body fat distribution, provides valuable information for developing personalized weight management tools and content. Finally, the insights into heart rate patterns can be used to inform more advanced features related to cardiovascular health and personalized training recommendations. By leveraging these data-driven insights, the business can enhance user engagement through

personalized goal setting, targeted interventions to reduce sedentary behavior, and the development of integrated features that consider the interplay between different health metrics. This comprehensive understanding of user behavior allows for the creation of a more effective and engaging fitness platform that ultimately empowers users to make healthier choices and improve their overall well-being.

Conclusion

Based on the analysis, we can conclude that most users typically walk between 3,800 to 10,742 steps daily, with peak activity consistently occurring on Saturdays and Tuesdays. Activities are dominated by "Lightly Active Minutes," which average around 200 minutes per day, indicating that the user base primarily engages in low-intensity movement rather than structured, high-intensity workouts like "Very Active Minutes," which remain critically low at an average of 20 minutes.

Sedentary behavior dominates the majority of user time, with most participants spending over 1,000 minutes per day in an inactive state. There is a strong positive correlation (0.99) between total steps and total distance, suggesting a direct relationship where increased steps correspond to greater distances covered. Furthermore, calorie expenditure follows a clear bimodal trend, peaking significantly in the evening between 17:00 and 19:00. Overall, the data highlights varying activity levels among users, emphasizing the prevalence of sedentary routines and a significant opportunity for the business to implement "Active Break" features to shift user focus toward more holistic health management.

STRAVA FITNESS

Power BI Report

Github: Strava_Fitness

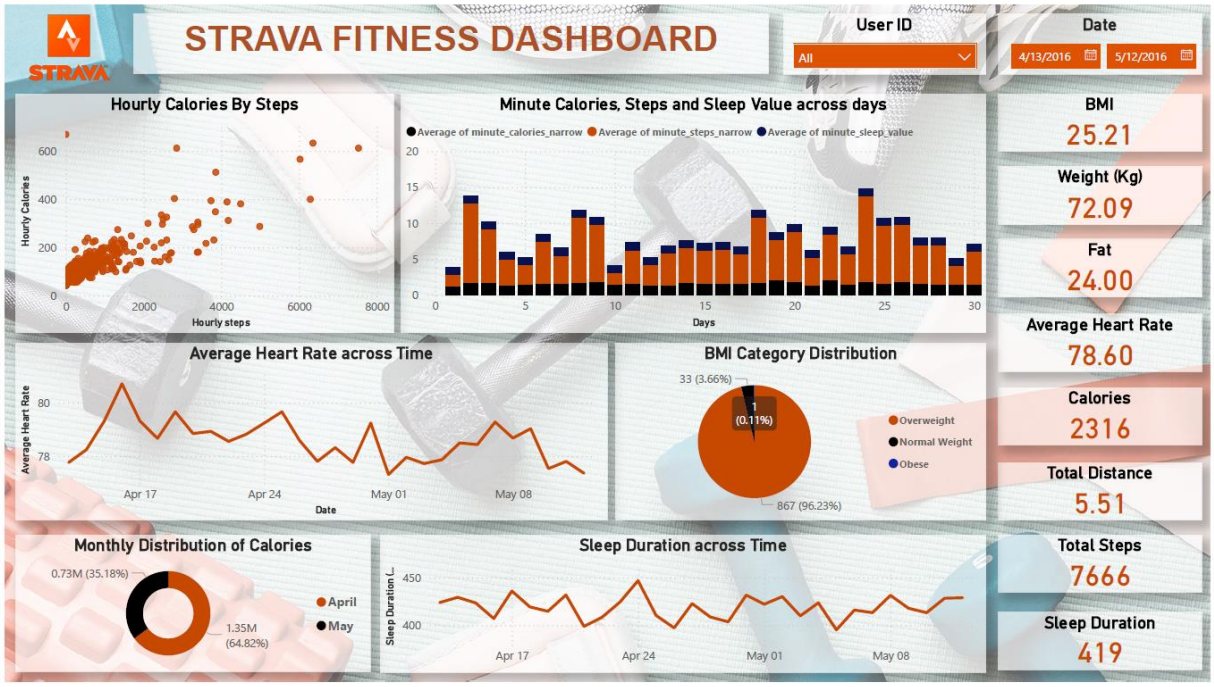
INTRODUCTION

This Power BI dashboard analyzes Strava fitness data for a one-month period (April 13 to May 12, 2016). It provides a holistic view of user health by integrating activity metrics, physiological data (BMI, weight, heart rate), and sleep patterns to understand overall wellness trends.

OBJECTIVE

The primary goal of this dashboard is to visualize the correlation between daily activities (steps, calories) and health outcomes (BMI, heart rate, sleep). It aims to identify patterns in user behavior and physiological health to provide actionable fitness insights.

STRAVA FITNESS DASHBOARD



DAX MEASURES USED

Based on the visual elements, the following core DAX measures were utilized:

- **Average Metrics:** Average Heart Rate (183), BMI (29.33), and Weight (40.21).
- **Total Aggregations:** Total Distance (10.06km), Total Steps (41,709K), and Total Minutes Calories (247K).

- **Percentage of Total:** Used in the BMI Category Distribution and Monthly Distribution of Calories.

KEY PERFORMANCE INDICATORS (KPIs)

The dashboard prominently tracks several health and fitness KPIs:

- **BMI:** 25.21 (Overweight)
- **Weight (Kg):** 72.09
- **Fat:** 24.00
- **Average Heart Rate:** 78.60 BPM
- **Calories:** 2,316
- **Total Distance:** 5.51
- **Total Steps:** 7,666
- **Sleep Duration:** 419 Minutes (~7 hours)

DATA VISUALIZATIONS

- **Hourly Calories By Steps (Scatter Plot):** Displays a strong positive correlation between hourly steps and calories burned. Most data points cluster between 0–2,000 steps, while high-intensity outliers reach over 600 calories.
- **Minute Calories, Steps and Sleep Value (Stacked Column Chart):** Tracks the daily volume of activity across 30 days. It reveals fluctuations in daily consistency, showing peaks and troughs in calorie expenditure and step counts.
- **Average Heart Rate across Time (Line Chart):** Monitors the daily average heart rate, showing a range between ~77 and ~80 BPM. This indicates a relatively stable resting/low-activity heart rate over the month.
- **BMI Category Distribution (Pie Chart):** A critical health metric showing that 96.23% (867 instances) of the data falls into the Overweight category, with only 3.66% (33 instances) in the Normal Weight category.
- **Monthly Distribution of Calories (Donut Chart):** Breaks down activity by month. April (64.82% / 1.35M calories) saw nearly double the engagement and calorie burn compared to May (35.18% / 0.73M calories).
- **Sleep Duration across Time (Line Chart):** Visualizes sleep patterns over the 30-day period. It shows significant variability, with the average duration sitting at 419 minutes (approximately 7 hours).

SLICERS

To allow for interactive data exploration, the dashboard includes:

- **Date Range Slicer:** Set from 4/13/2016 to 5/12/2016.
- **ID Slicer:** Allows for filtering the entire report by specific individual participants.

BUSINESS INSIGHTS SUMMARY

- **Health Profile:** The overwhelming majority (**96.23%**) of users are in the "Overweight" BMI category (25.21), suggesting a target market for weight-loss-focused features.
- **Activity Gap:** The **Total Steps (7,666)** fall short of the universally recommended 10,000-step goal for optimal health.
- **Retention Drop:** There is a significant decline in activity volume from **April to May**, dropping from 64.82% to 35.18% of total monthly calories.

Business Recommendations

- **Personalized Coaching:** Introduce a "Road to 10k" step challenge to help users bridge the gap from 7,666 to 10,000 steps.
- **Weight Management Tools:** Since BMI is high (25.21), integrate a calorie-deficit tracker or nutritional logging to complement the physical activity data.
- **Engagement Re-ignition:** Implement push notifications or mid-month "milestone" rewards to prevent the type of engagement drop seen in May.

BUSINESS RECOMMENDATIONS

- **Targeted Weight Management:** Develop specialized "Step Challenges" for users in the higher BMI categories to encourage gradual weight loss.
- **Heart Rate Monitoring:** Integrate "Recovery Zone" alerts for users maintaining high average heart rates to ensure they are resting sufficiently.
- **Sleep Hygiene:** Given the variability in sleep duration, introduce in-app prompts for consistent sleep schedules to improve overall recovery.

CONCLUSION

The Strava Fitness Dashboard effectively highlights that while users are achieving high step counts and calorie burns, their physiological markers (BMI and Heart Rate) suggest a need

for more balanced health coaching. By focusing on consistency and recovery, the business can move beyond simple tracking to becoming a true health improvement partner. The dashboard reveals a user base that is moderately active but faces specific health risks related to BMI. By leveraging the clear correlation between steps and calories shown in the scatter plot, Strava can drive better health outcomes through increased daily movement targets and consistent sleep tracking.